NORMAL WAVEFORMS, ARTIFACTS, AND PITFALLS

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The proper collection and interpretation of hemodynamic waveforms are important components of cardiac catheterization, yet they are often overshadowed by the more glamorous aspects of cardiology, such as angiography and coronary or structural intervention. Accurate intracardiac pressure measurements provide invaluable physiological information in both normal and disease states; poorly gathered or erroneously interpreted waveforms may lead to an incorrect diagnosis or a wrong clinical decision. It is imperative that competent cardiologists understand normal and abnormal hemodynamic waveforms and can troubleshoot problems. It is also important to recognize common artifacts and understand potential pitfalls and certain unique circumstances that may occur during the collection and interpretation of hemodynamic data.

GENERATION OF PRESSURE WAVEFORMS

The goal of a hemodynamic study is to accurately reproduce and analyze the changes in pressure that occur within a cardiac chamber during the cardiac cycle. These rapidly occurring events represent mechanical forces and require conversion to an electrical signal to be transmitted and subsequently translated into an interpretable graphic format. The pressure transducer is the essential component that translates mechanical forces to electrical signals. The transducer may be located at the tip of the catheter (micromanometer) within the chamber or, more commonly, the pressure transducer is outside the body, and a pressure waveform is transmitted from the catheter tip to the transducer through long connecting tubing containing a column of saline fluid. These transducers consist of a diaphragm or membrane attached to a strain gauge-Wheatstone bridge arrangement. When a fluid wave strikes the diaphragm, an electrical current is generated with a magnitude dependent on the strength of the force that deflects the membrane. The output current is amplified and displayed as pressure versus time.

During the cardiac cycle, changes in pressure occur rapidly, corresponding with various physiological events. The force wave created by these events generates a spectrum of wave frequencies. Consider, for example, the different events that occur within the aorta throughout a single cardiac cycle. Beginning with left ventricular ejection, the aortic valve opens, causing pressure to steadily rise in the aorta, rapidly reaching a peak, and then falling quickly following peak ejection. Closure of the aortic valve represents another event, marking the end of systole and is associated with a slight rise in pressure during the pressure decay phase and then followed by continued pressure decay during ventricular diastole until the next ventricular contraction. All these events occur rapidly and are associated with varying wave frequencies. When the force wave strikes the transducer, it should precisely reproduce these events. The full range of frequencies needed to reproduce these force waves requires a sensing membrane capable of a rapid frequency response (0–20 cycles/s in the human heart). However, the physical properties of a membrane capable of such a wide frequency response might resonate, creating artifacts. This phenomenon is similar to the sound that a bell makes, continuing to oscillate after initially struck. Resonation artifact appears on the waveform as excessive "noise," but reverberations can also lead to harmonic amplification of the waveform overestimating systolic pressure and underestimating diastolic pressure (Fig. 2.1). Therefore hemodynamic measurement systems need to provide some level of "damping" to reduce resonation artifacts. Damping is a method of eliminating the oscillation; it may be done by the introduction of friction to reduce the oscillation of the sensing membrane, or it may be accomplished electrically by damping algorithms. However, note that damping reduces the frequency response and thus may result in loss of information. Overdamping results in the loss of rapid high-frequency events (e.g., the dicrotic notch on the aortic waveform), causing underestimation of the systolic pressure and overestimation of the diastolic pressure. The ideal pressure tracing has the proper balance of frequency response and damping.

CALIBRATION, BALANCING, AND ZEROING

Previous generations of transducers required calibration against a mercury manometer; this is no longer needed with the factory-calibrated, disposable, fluid-filled transducers in clinical use today. Table-mounted transducers do require balancing or "zeroing," which refers to the establishment of a reference point for subsequent pressure measurements. The reference or "zero" position should be determined before any measurements are made. By convention, it is defined at the patient's midchest in the anteroposterior dimension at the level of the sternal angle of Louis (fourth intercostal space) (Fig. 2.2). This site is an estimation of the location of the right atrium and is also known as the *phlebostatic axis*. A table-mounted transducer is placed at this level, and the stopcock is opened to air (atmospheric pressure) and set to zero by the hemodynamic system. The system is now ready for pressure measurements.

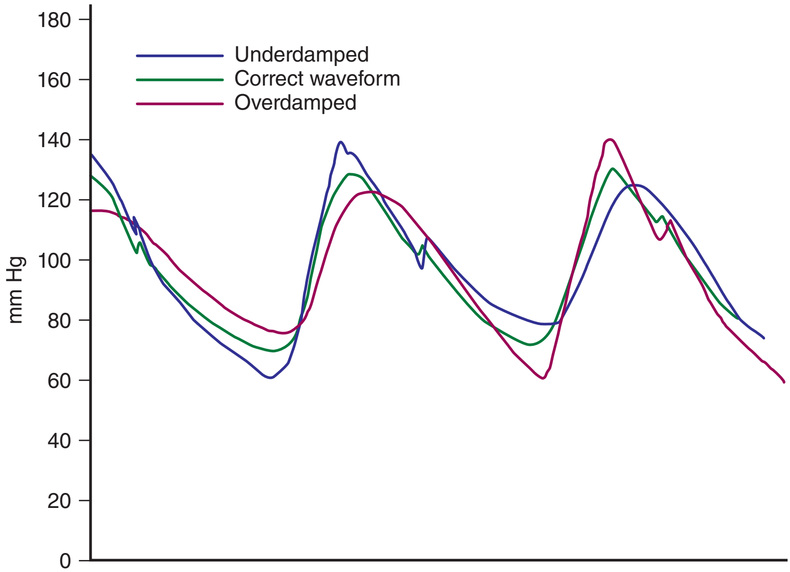


Fig. 2.1 Schematic representation of the effects of overdamping or underdamping on the pressure waveform. An underdamped waveform will overestimate systolic pressure and underestimate diastolic pressure, whereas overdamping will have the opposite effects. In addition, the overdamped waveform obscures subtle hemodynamic findings, such as the dicrotic notch.

The tradition of using the midaxillary line as the zero position has been called into question. Because of the influence of hydrostatic pressure in the supine position, and the use of fluid-filled transducers, some physicians believe that setting the zero position as the upper border of the left ventricle is more accurate.[1] The difference between this location and the conventional location provides the greatest accuracy in diastolic pressure measurements. However, most routine laboratories find this approach impractical because it requires the use of echocardiography to determine the precise location; it is more applicable to research investigations. A major advantage of the midchest position is that it has been shown to correlate with the position of the left atrium by magnetic resonance imaging studies, regardless of the patient's age, sex, body habitus, or presence of chronic lung disease.[2] Frequently, busy catheterization laboratories might position the transducer at the same level for all patients, or from a measured, fixed distance from either the table or from the top of the patient's chest, without taking into consideration the variations in patient position or body habitus. This practice will lead to marked inaccuracies, particularly in patients who are unable to lie flat or who are at the extremes of body weight. A transducer placed *above* the true zero position will produce a pressure *lower* than the actual pressure; a transducer placed *below* the true zero position will result in a pressure measurement *higher* than the actual pressure. These small pressure changes caused by improper zeroing will be more impactful in the lower, diastolic heart pressures and may lead to significant errors in diagnosis and, perhaps, inappropriate therapy.

Transducer *drift* refers to either the loss of calibration or the loss of balance after initially setting the zero level. This is not uncommon. Many patients have been started on pressors for hypotension or have been falsely diagnosed with mitral stenosis because of inaccurate transducer balancing, improper zero positioning, or transducer drifting. Careful attention to this aspect is important for proper interpretation. When in doubt, if the findings do not make sense with the clinical scenario, or if the result is needed for an important decision, recheck the zero.

NORMAL PHYSIOLOGY AND WAVEFORM CHARACTERISTICS

Interpretation of pressure waveforms requires a consistent and systematic approach (Box 2.1). After confirming the zero level, the scale of the recording is noted, and a recording sweep speed is determined. Establishing standard protocols is helpful to ensure that all necessary information is collected in a systematic format (see Chapter 1). Careful scrutiny of the waveform ensures a high-fidelity recording without overdamping or underdamping. Each pressure event should be timed with the electrocardiogram (ECG). Finally, the operator should review the tracings for the presence of common artifacts that might lead to misinterpretation.

The various events occurring during the cardiac cycle create characteristic waveform appearances. Fig. 2.3 demonstrates the three basic waveforms (atrial, ventricular, and arterial) and their relationships to the key electrical and physiological events during the normal cardiac cycle. Each waveform will be described in detail in the following sections.

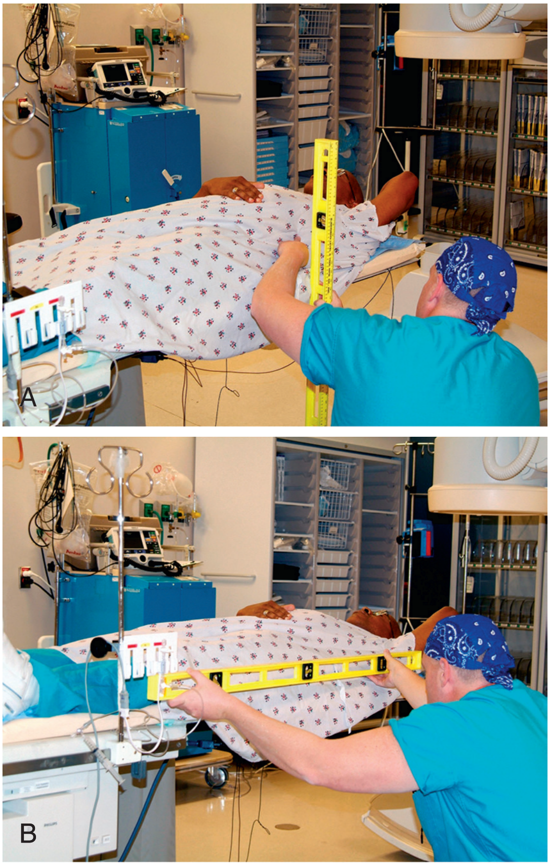


Fig. 2.2 (A) Demonstration of the "zero" position or phlebostatic axis, representing a point midway in the anteroposterior chest dimension at the fourth intercostal space. (B) Table-mounted transducers are positioned at this point using a level to ensure accuracy.

Box 2.1 A Systematic Approach to Hemodynamic Interpretation

1. Establish the zero level and balance transducer
2. Confirm the scale of the recording
3. Collect hemodynamics in a systematic method using established protocols
4. Critically assess the pressure waveforms for proper fidelity
5. Carefully time pressure events with the electrocardiogram
6. Review the tracings for common artifacts

Right Atrial Waveform

The normal right atrial pressure is 2 to 6 mm Hg and is characterized by *a* and *v* waves and *x* and *y* descents (Fig. 2.4). The *a* wave represents the pressure rise within the right atrium due to atrial contraction and follows the *P* wave on the ECG by about 80 ms. The *x* descent represents the pressure decay following the *a* wave and reflects both atrial relaxation and the sudden downward motion of the atrioventricular (AV) junction that occurs because of ventricular systole. An *a* wave is usually absent in atrial fibrillation, but the *x* descent may be present because of this latter phenomenon (Fig. 2.5). A *c* wave is sometimes observed after the *a* wave and is due to the sudden motion of the tricuspid annulus toward the right atrium at the onset of ventricular systole. The *c* wave follows the *a* wave at the same time as the PR interval on the ECG; thus first-degree AV block results in a more obvious *c* wave (Fig. 2.6). When a *c* wave is present, the pressure decay following it is called an x^1 descent.

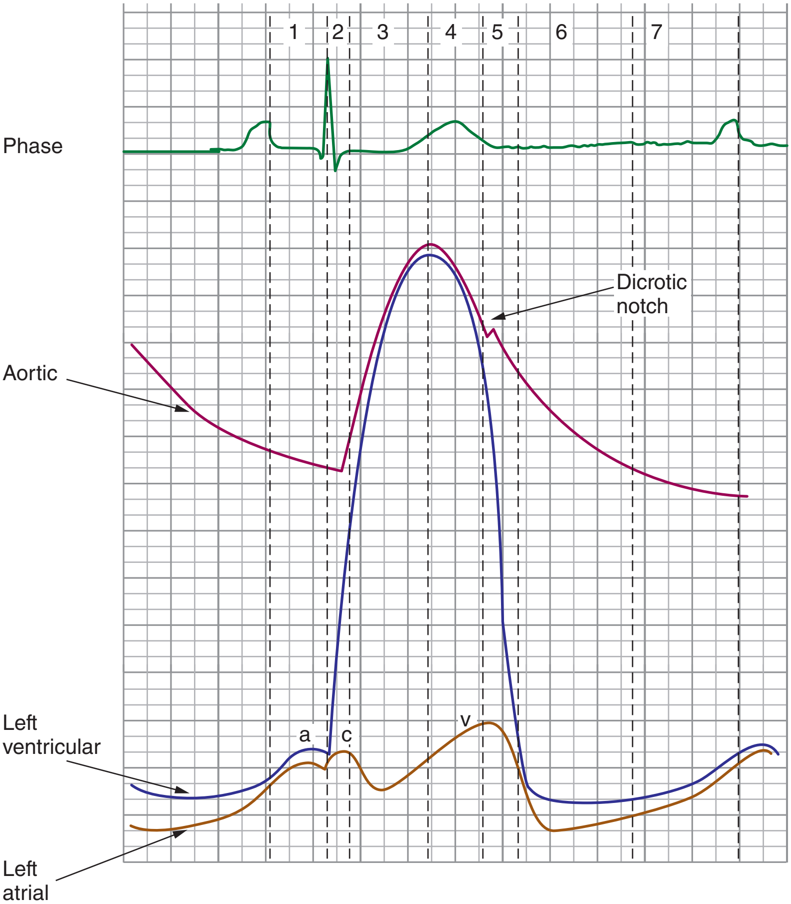


Fig. 2.3 Timing of the major electrical and mechanical events during the cardiac cycle. Phase 1, atrial contraction; phase 2, isovolumic contraction; phase 3, rapid ejection; phase 4, reduced ejection; phase 5, isovolumic relaxation; phase 6, rapid ventricular filling; phase 7, reduced ventricular filling.

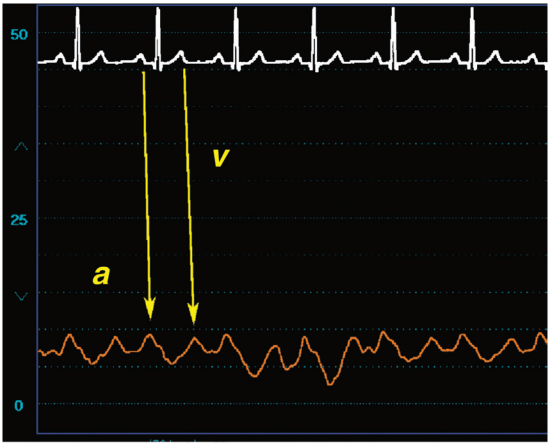


Fig. 2.4 An example of a normal right atrial pressure waveform. Note the timing from the electrocardiographic *P* and *T* waves to the hemodynamic *a* and *v* waves, respectively.

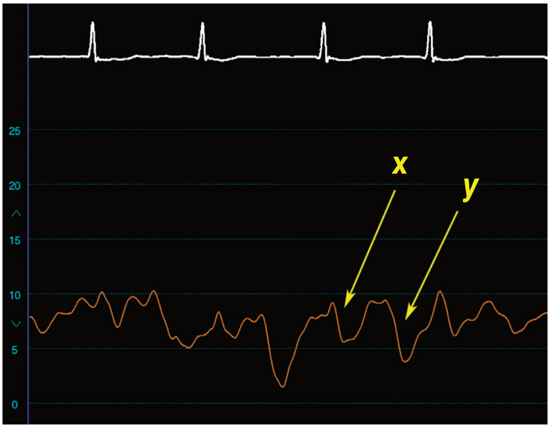


Fig. 2.5 Right atrial waveform obtained in a patient with chronic atrial fibrillation, demonstrating persistence of the *x* descent despite loss of the *a* wave.

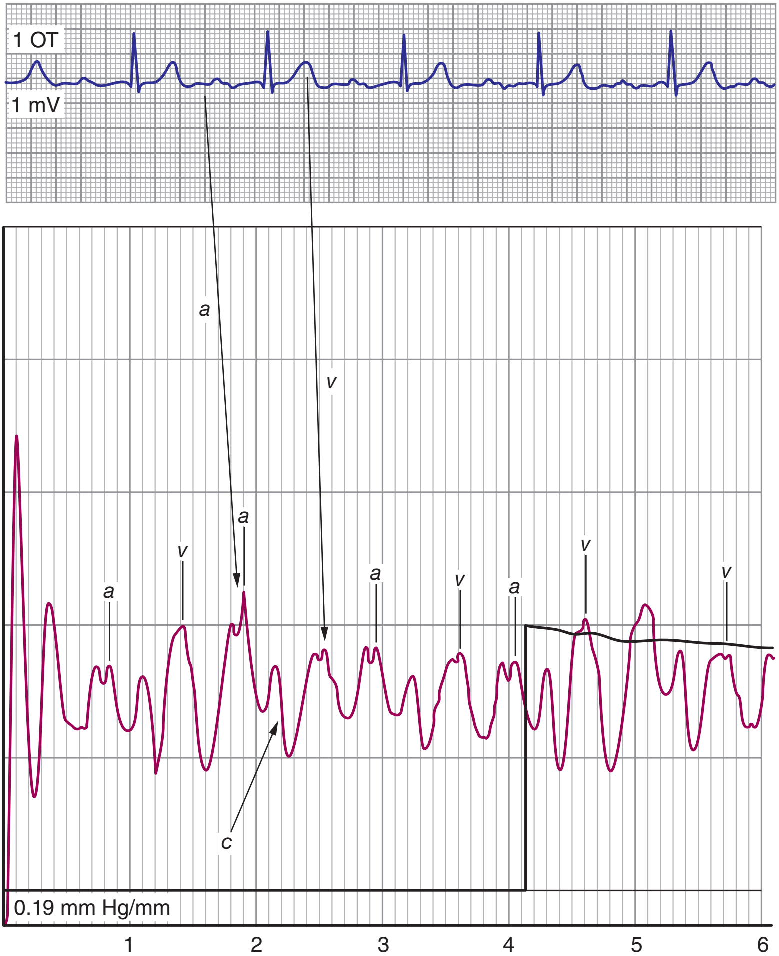


Fig. 2.6 Example of a right atrial waveform with prominence of the *c* wave due to the presence of first-degree atrioventricular block.

The next pressure event is the *v* wave. There is often misunderstanding regarding the mechanism of the *v* wave. Although this event occurs at the same time as ventricular systole (and when the tricuspid valve is closed), the pressure rise responsible for the *v* wave is not due to ventricular systole but to passive venous filling of the atrium during atrial diastole. Conditions that increase filling of the right atrium create greater prominence of the *v* wave. The peak of the right atrial *v* wave occurs at the end of ventricular systole, when the atria are maximally filled, and corresponds with the end of the *T* wave on the surface ECG. The pressure decay that occurs after the *v* wave is the *y* descent and is due to the rapid emptying of the right atrium when the tricuspid valve opens. Atrial contraction follows this event and the onset of another cardiac cycle. In normal right atrial waveforms, the *a* wave typically exceeds the *v* wave. During inspiration the mean right atrial pressure decreases due to the influence of decreased intrathoracic pressure, and there is an augmentation of passive right ventricular filling; the *y* descents become more prominent (Fig. 2.7).

Right Ventricular Waveform

The normal right ventricular systolic pressure is 20 to 30 mm Hg, and the normal right ventricular end-diastolic pressure is 0 to 8 mm Hg. Right ventricular tracings exhibit the characteristic features of ventricular waveforms with rapid pressure rise during ventricular contraction and rapid pressure decay during relaxation, with a diastolic phase characterized by an initially low pressure that gradually increases (Fig. 2.8). The right atrial pressure should be within a few millimeters of mercury of right ventricular end-diastolic pressure unless there is tricuspid stenosis. With atrial contraction, an *a* wave may appear on the ventricular waveform at end diastole (Fig. 2.9), which is not a normal finding because the normal, compliant right ventricle typically absorbs the atrial component without a significant pressure rise. Therefore the presence of *a* wave on a right ventricular waveform usually indicates decreased compliance as may be seen in patients with pulmonary hypertension, right ventricular hypertrophy, or volume overload.

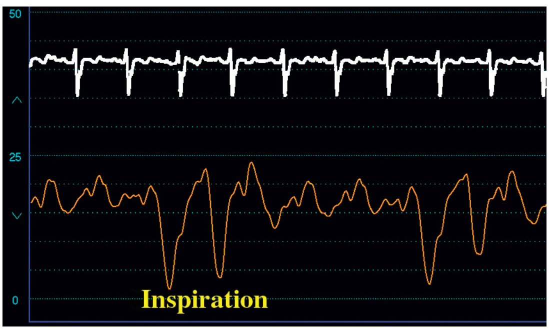


Fig. 2.7 With inspiration, the *x* and *y* descents become more prominent on the right atrial waveform.

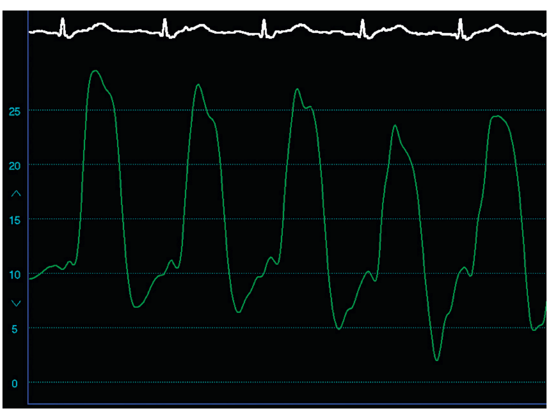


Fig. 2.8 Example of a normal right ventricular waveform.

Pulmonary Artery Waveform

The normal pulmonary artery systolic pressure is 20 to 30 mm Hg, and the normal diastolic pressure is 4 to 12 mm Hg (Fig. 2.10). A systolic pressure difference should not exist between the right ventricle and the pulmonary artery, unless there is pulmonary valvular or pulmonary artery stenosis. The pulmonary artery pressure tracing is similar to other arterial waveforms, with a rapid rise in pressure, systolic peak, a pressure decay following peak ejection, and a well-defined dicrotic notch from pulmonic valve closure during the pressure decay culminating in a diastolic trough. Peak systolic pressure occurs at the same time as the *T* wave on the surface ECG.

The pulmonary artery waveform, like other right heart chamber pressure waveforms, is subject to respiratory changes (Fig. 2.11). Inspiration decreases intrathoracic pressure, and expiration increases intrathoracic pressure. The pressure changes associated with respiration transmitted to the cardiac chambers are often small and of little consequence. However, patients on mechanical ventilators with severe pulmonary disease, morbid obesity, or respiratory distress may generate substantial changes in intrathoracic pressure, resulting in marked differences in pulmonary artery pressures during the respiratory phases (Fig. 2.12). Most experts consider end expiration to be the proper point to assess pulmonary artery (and other cardiac chamber) pressures because it is at this phase that intrathoracic pressure is closest to zero.[3] If this effect is prominent and makes interpretation difficult, many operators will ask the patient to pause their breathing at end expiration during pressure sampling. The pulmonary artery end-diastolic pressure is sometimes used as an estimation of the left atrial pressure; however, it is highly inaccurate, especially if the pulmonary vascular resistance is abnormal.[4]

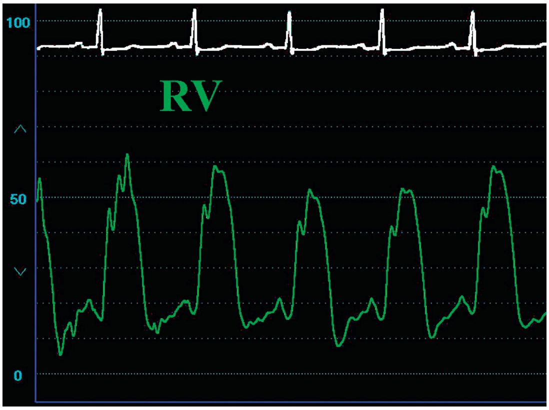


Fig. 2.9 Right ventricular (RV) waveform obtained in a patient with pulmonary hypertension and RV hypertrophy, demonstrating prominent *a* waves.

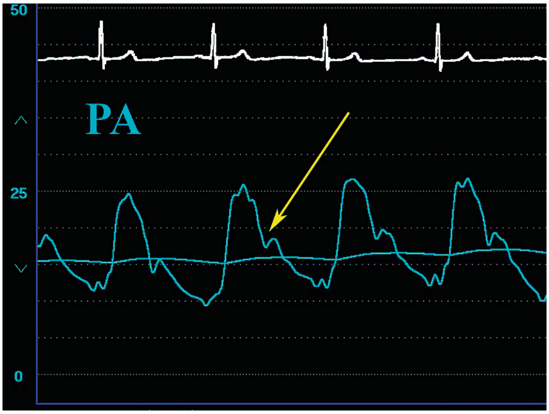


Fig. 2.10 An example of a normal pulmonary artery (PA) waveform. Note the dicrotic notch *(arrow).*

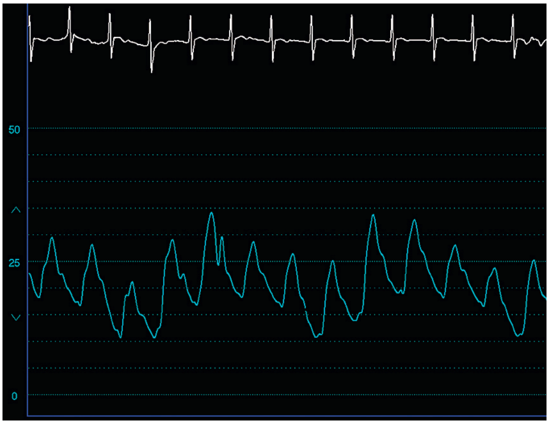


Fig. 2.11 Respiratory variation in a pulmonary artery pressure wave.

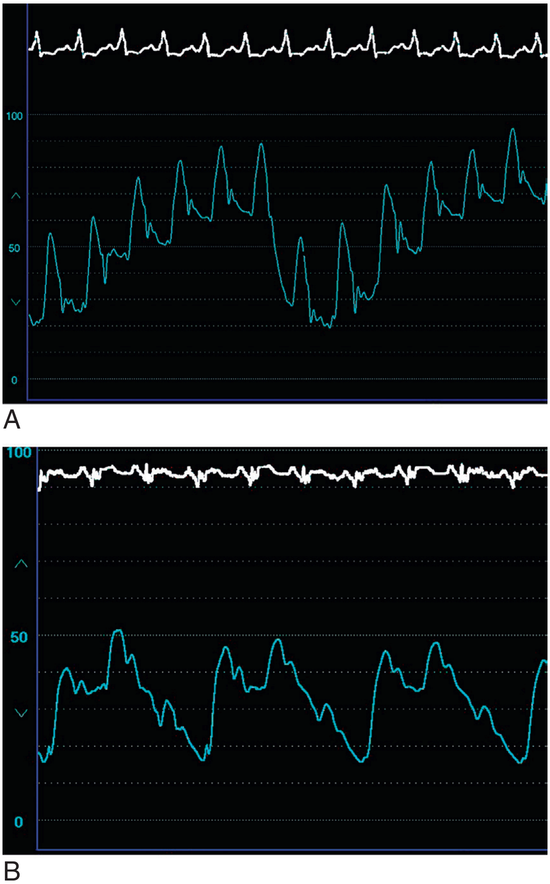


Fig. 2.12 Marked respiratory variation in pulmonary artery pressure. (A) A patient with marked pulmonary hypertension. (B) A patient with morbid obesity.

Novel Methods of Measuring Pulmonary Artery Pressure in the Patient With Heart Failure

Invasive hemodynamic assessments are immensely helpful in understanding cardiac disease processes and targeting appropriate therapy. However, these measurements are limited to a single “snapshot” in the cardiac catheterization laboratory, or occasionally, several days of monitoring in an intensive care unit. These may not accurately reflect the patient’s hemodynamic status at home under normal activities. Technology to allow long-term hemodynamic monitoring has been in development for years, including left atrial, right ventricular, and pulmonary artery pressure sensors. Currently, there is approval and clinical use of the CardioMEMS (Abbott, Chicago, Illinois) pulmonary artery pressure sensor.[5] This device is implanted into a branch of a pulmonary artery (Fig. 2.13) and calibrated by simultaneous invasive measurements. Subsequent measurements are obtained from the pulmonary artery pressure sensor by an external receiver, generating a pulmonary artery waveform (Fig. 2.14), which is transmitted to the care team for review and intervention. The interpretation software allows the review of pulmonary artery pressure trends over time (Fig. 2.15). Based on the results of several clinical trials, the device is approved in the United States and is indicated to reduce heart failure hospitalizations in patients with New York Heart Association Class II–III symptoms with prior hospitalization for heart failure or elevation of brain natriuretic peptide.[6–8] Heart rate and arrhythmia information is easily obtained from the waveforms by manual inspection, and algorithms to monitor cardiac output are in development.

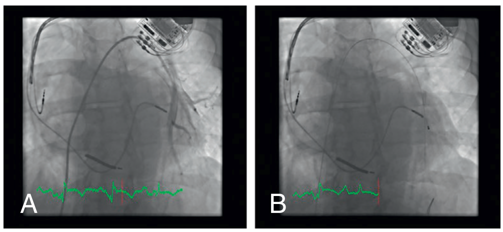


Fig. 2.13 Insertion of the CardioMEMS device in the pulmonary artery. (A) An angiogram of the left pulmonary artery. (B) The device implanted into a branch of the left pulmonary artery.

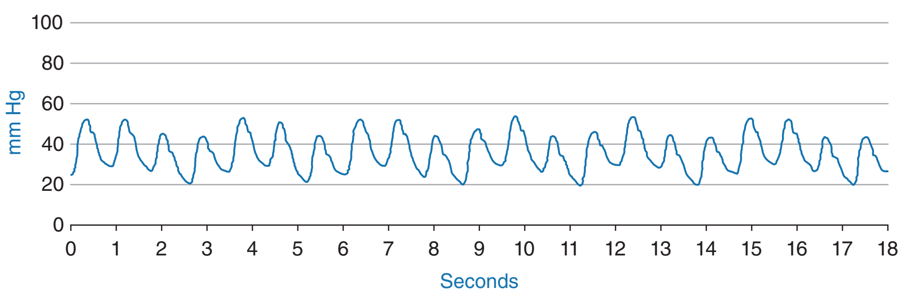


Fig. 2.14 This is an example of the pulmonary artery waveform obtained from interrogation of the CardioMEMS device. (Courtesy CardioMEMS; Abbott, Chicago, Illinois.)

Fig. 2.15 The CardioMEMS device can provide the clinician with trends in pulmonary artery (PA) pressure over time, as shown in this example. (Courtesy CardioMEMS; Abbott, Chicago, Illinois.)

Pulmonary Capillary Wedge Pressure Waveform

The normal mean pulmonary capillary wedge pressure (PCWP) is 2 to 14 mm Hg (Fig. 2.16). A true PCWP can be measured only in the absence of anterograde flow in the pulmonary artery and with an end-hole catheter, such that pressure is transmitted through an uninterrupted fluid column from the left atrium, through the pulmonary veins and pulmonary capillary bed, to the catheter tip wedged in the pulmonary artery. Under these circumstances, the PCWP is a reflection of the left atrial pressure with *a* and *v* waves and *x* and *y* descents.

The PCWP tracing exhibits several important differences from a directly measured atrial pressure waveform. The *c* wave is absent because of the damped nature of the pressure wave. The *v* wave typically exceeds the *a* wave on the PCWP tracing. Because the pressure wave is transmitted through the pulmonary capillary bed, a significant time delay occurs between an electrocardiographic event and the onset of the corresponding pressure wave. The delay may vary substantially, depending on the distance the pressure wave travels. Shorter delays are observed when the PCWP is obtained with the catheter tip in a more distal location. Typically, the peak of the *a* wave follows the *P* wave on the ECG by about 240 ms, rather than 80 ms as seen in the right atrial tracing.[9] Similarly, the peak of the *v* wave occurs after the *T* wave has already been inscribed on the ECG. The relationship between true left atrial pressure and PCWP is shown in Fig. 2.17. Note the time delay between the same physiological events and the "damped" nature of the PCWP relative to the left atrial waveform, with a pressure slightly lower than the left atrium it is meant to reflect. In general, the mean PCWP is within a few millimeters of mercury of the mean left atrial pressure, especially if the wedge and pulmonary artery systolic pressures are low.[10] High pulmonary artery pressure creates difficulty in obtaining a true "wedge," falsely elevating the PCWP relative to the left atrial pressure.

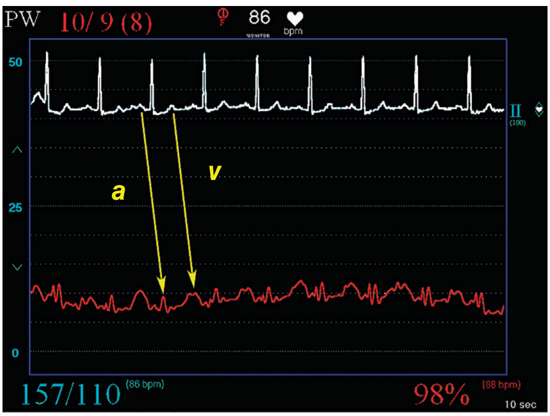


Fig. 2.16 A normal pulmonary capillary wedge pressure waveform with distinct *a* and *v* waves.

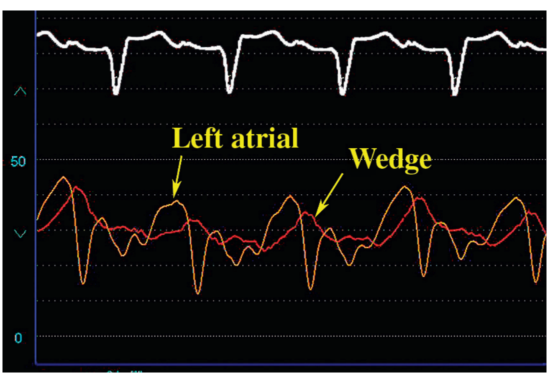


Fig. 2.17 Relationship between the left atrial and pulmonary capillary wedge pressure (PCWP) waveforms. Note the time delay on the PCWP for the same events and the relatively "damped" appearance of the PCWP tracing with a slightly lower pressure compared with the left atrial pressure.

Obtaining an accurate and high-quality PCWP tracing is not always easy or possible. An uninterrupted fluid column between the catheter tip and the left atrium is important. However, the lung consists of three distinct physiological pressure zones, with a different relationship between the alveolar, pulmonary artery, and pulmonary venous pressures (the lung zones of West) (Fig. 2.18).[11] Zone 1 is typically present in the apex of the lungs, where the alveolar pressure is greater than the mean pulmonary artery and pulmonary venous pressures. Zone 2 is located in the central portion of the lung, and pulmonary artery pressure exceeds alveolar pressure, which, in turn, is greater than the pulmonary venous pressure. These zones are not acceptable for estimation of the PCWP because capillary collapse is present based on these pressure relations, and a direct column of blood does not exist between the left atrium and the wedged catheter tip. Lung zone 3 is represented by the base of the lung, where alveolar pressure is lower than both pulmonary arterial and pulmonary venous pressure, allowing pressure transmission directly from the left atrium to the wedged catheter tip. Lung zone 3 is where PCWP accurately reflects the left atrial pressure. Fortunately, in most patients in the supine position on a cardiac catheterization table, most of the lung is in zone 3. In addition, because most blood flows to that area, the catheter tip of a balloon flotation catheter usually ends up in zone 3. Situations associated with catheter tip location in a nonzone 3 location include the use of positive end-expiratory pressure (PEEP), mechanical ventilation (alveolar pressure is increased and less of the lung is zone 3), and hypovolemia. Demonstrating that the catheter tip is below the level of the left atrium, however, ensures a zone 3 location and greater accuracy.[3]

Characteristics of a high-quality PCWP include (1) the presence of well-defined *a* and *v* waves (note that the *a* wave is absent in atrial fibrillation, and phasic waves may not be distinct at low pressures); (2) appropriate fluoroscopic confirmation with the catheter tip in the distal pulmonary artery and no apparent motion of the catheter with the balloon inflated; (3) an oxygen saturation obtained from the PCWP position greater than 90%; and (4) observation of a distinct, abrupt rise in mean pressure when the balloon is deflated or the catheter is withdrawn from the PCWP position to the pulmonary artery. Of all these signs, obtaining an oxygen saturation greater than 90% from the catheter tip is the most confirmatory of a true PCWP. An "overwedged" pressure occurs when the catheter tip is in a peripheral pulmonary artery and the balloon is overinflated; this catheter position may lead to pulmonary artery rupture, a potentially fatal complication of pulmonary artery catheterization. The overwedged tracing is a false PCWP measurement and appears as a wavering line without distinct *a* and *v* waves and does not reflect the left atrial pressure.

The mean PCWP is approximately 0 to 5 mm Hg lower than the pulmonary artery diastolic pressure, unless there is increased pulmonary vascular resistance. Obtaining a suitable and accurate wedge pressure may be difficult or impossible for patients with pulmonary hypertension. Accordingly, if it is important to measure the left atrial pressure accurately, such as during the evaluation of mitral stenosis, and if the operator is unable to confirm the wedge pressure, then a transseptal catheterization with direct measurement of the left atrial pressure is necessary. Similar to other right-sided pressure tracings, the end-expiratory wedge pressure is most representative of the true hemodynamic status if there is a great deal of respiratory variation.

Overall, a good correlation exists between the pulmonary capillary wedge, left atrial, and left ventricular end-diastolic pressures (LVEDPs). The PCWP does not correlate with LVEDPs when there is mitral stenosis, severe mitral or aortic regurgitation, pulmonary venous obstruction, marked increase in PEEP, left atrial myxoma, marked left ventricular noncompliance, or nonzone 3 location of the catheter tip.[12] In patients with large *v* waves, the trough of the *x* descent is the best predictor of the LVEDP.[13]

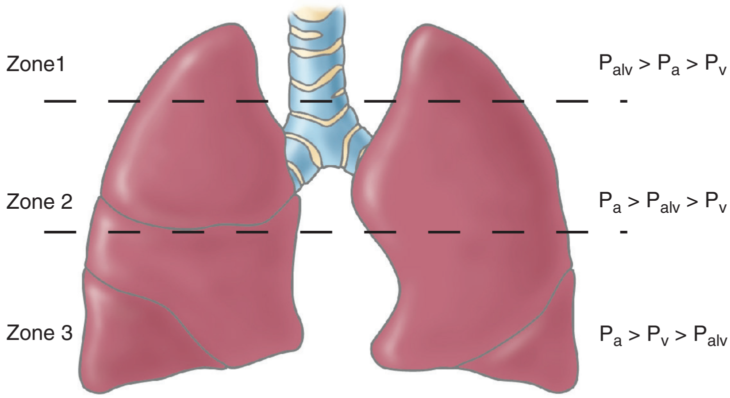


Fig. 2.18 Schematic representation of the three lung zones of West. A true wedge pressure can be obtained only when an uninterrupted column of blood exists from the pulmonary vein to the pulmonary artery. In zone 1, alveolar pressure is the highest pressure, compressing both vessels; in zone 2, although pulmonary arterial pressure exceeds pulmonary alveolar pressure, the pulmonary venous system is compressed by the higher alveolar pressures. Zone 3 is the only area where alveolar pressure is lower than pulmonary venous pressure and does not interfere with the column of blood, thus allowing an accurate wedge pressure. P_a, Pulmonary artery pressure; P_{alv}, alveolar pressure; P_v, pulmonary venous pressure.

Determination of the PCWP in patients on a ventilator with PEEP poses a common clinical dilemma. PEEP increases alveolar pressure, reducing the proportion of lung zone 3. In addition, the positive pressure is transmitted to the central circulation, directly affecting right-sided pressures and leading to an overestimation of the PCWP. The extent to which PEEP increases right-sided chamber pressures is not predictable and depends on such variables as compliance of the cardiac chamber, chest wall and lung, volume status, and existing filling pressures. The general consensus is that PEEP less than 10 cm H_2O does not significantly affect the PCWP. Still, debate exists on methods to correct the PCWP when PEEP exceeds 10 cm H_2O. The effect of PEEP on intrathoracic pressure can be determined by subtracting the esophageal pressure from the PCWP, but this method is not practical in most coronary care units or cardiac catheterization laboratories. One suggested method for correction is based on the observation that PCWP rises 2 to 3 cm for every 5 cm H_2O increment in PEEP.[12]

Left Ventricular Waveform

The normal left ventricular systolic pressure is 90 to 140 mm Hg, and the normal end-diastolic pressure is 10 to 16 mm Hg. The left ventricular waveform is characterized by a very rapid upstroke during ventricular contraction, reaching a peak systolic pressure, and then the pressure rapidly decays (Fig. 2.19). The pressure in early diastole is typically very low and slowly rises during diastole. Abnormalities in ventricular relaxation are apparent when early diastolic pressure is high and declines during diastole.[14] Similar to the right ventricular waveform, an *a* wave may be seen in the left ventricular tracing at end diastole; however, this is usually abnormal and implies a noncompliant left ventricle. LVEDP is defined as the pressure just after the *a* wave and before the abrupt rise in systolic pressure coinciding with ventricular ejection. However, identification of the LVEDP can sometimes be difficult. A late diastolic pressure rise may become incorporated into the left ventricular upstroke, obscuring the LVEDP (Fig. 2.20). Again, in the presence of large and prominent *a* waves, the pressure just after the *a* wave represents the LVEDP (Fig. 2.21). Intrathoracic pressure changes due to the respiratory phases can also affect LVEDP, the pressure at end expiration (Fig. 2.22).

Central Aortic Pressure Waveform

The normal aortic systolic pressure is 90 to 140 mm Hg, and the normal diastolic pressure is 60 to 90 mm Hg. Central aortic waveforms have a rapid upstroke, a systolic peak, and a clearly defined dicrotic notch due to the closure of the aortic valve during pressure decay (Fig. 2.23). The peak systolic pressure equals the peak left ventricular systolic pressure, unless there is an obstruction within the left ventricle, at the aortic valve, or within the proximal aorta. An anacrotic notch may be apparent during the systolic pressure rise (Fig. 2.24) as a result of turbulent flow during ejection and indicates an abnormality in the aortic valve or proximal aorta. The anacrotic notch is commonly present in patients with severe aortic stenosis but may also be seen in patients with aortic regurgitation.

The measured central aortic pressure is composed of two components: the pressure wave generated from forward flow from left ventricular ejection, and the summation of pressure waves generated from “reflected” waves. Reflected waves result because as blood is ejected forward, it meets areas of resistance such as branch points or tortuous vessels. When the pressure wavefront strikes these areas, additional pressure waves are generated and directed back to the heart. These pressure waves strike the aortic valve, generating additional, forward-directed, smaller pressure waves. Therefore the sampled pressure waveform represents a summation of all the forward impulses.

This phenomenon is less apparent when pressure is sampled closer to the aortic valve. The effect increases the further away from the aortic valve that pressure is sampled. However, this effect is usually negligible and only raises the systolic pressure a small amount. Under some circumstances, however, the reflected waves may be of significant dimension. Factors that increase the effect of reflected waves include heart failure, aortic regurgitation, systemic

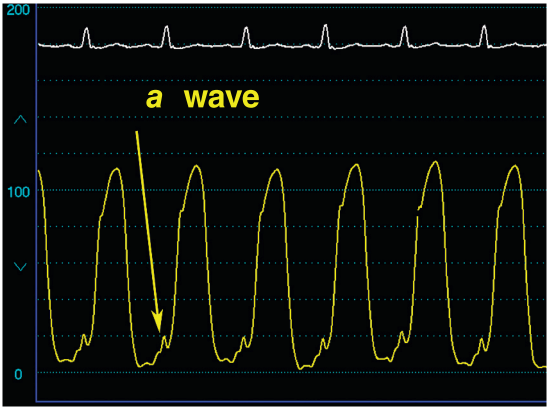


Fig. 2.19 An example of a high-fidelity left ventricular pressure waveform. An *a* wave is present on this tracing, which is not always seen on left ventricular pressure waves, unless diminished compliance of the left ventricle is present.

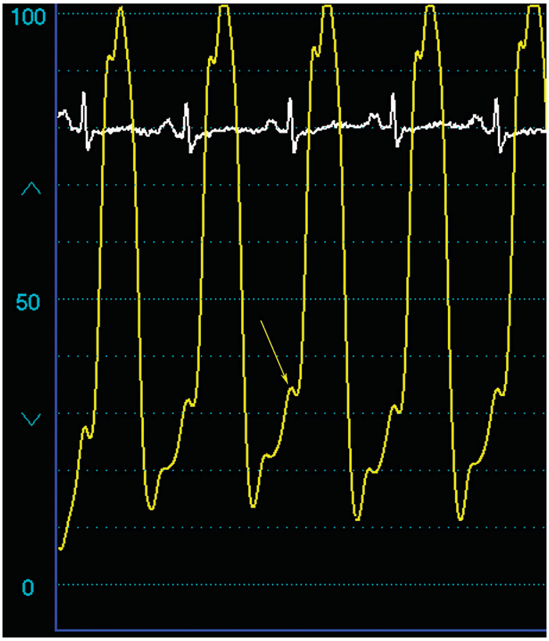


Fig. 2.20 Marked elevation of the left ventricular end-diastolic pressure *(arrow)* in a patient with heart failure.

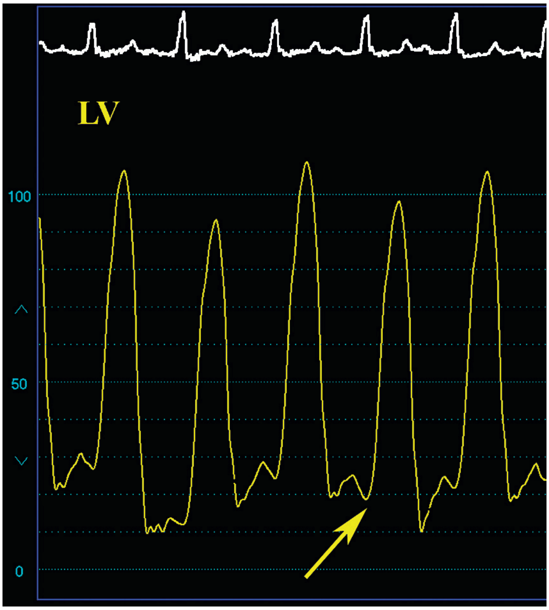


Fig. 2.21 Large *a* wave on left ventricular (LV) pressure wave. When a large *a* wave is present, the LV end-diastolic pressure is identified as the pressure just after the *a* wave and before the LV systolic pressure rise *(arrow)*.

hypertension, increased aortic stiffness from advanced age or peripheral vascular disease, aortic or iliofemoral obstruction, or tortuosity and arterial vasoconstriction. Factors associated with diminished reflected waves include vasodilation, hypovolemia, and hypotension. The reflected wave is typically apparent late in the aortic waveform due to the time delay from its generation to its summation with the forward waves.

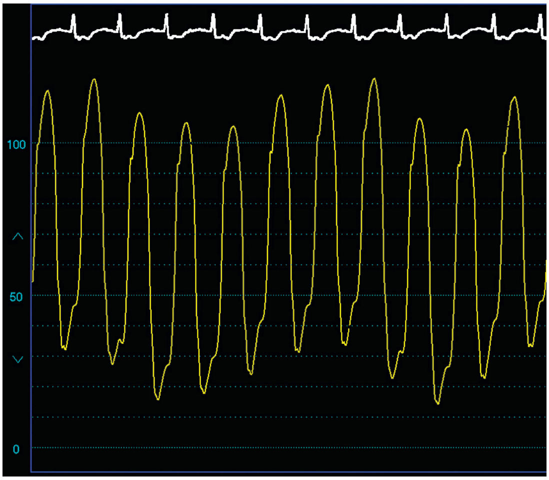


Fig. 2.22 Marked respiratory variation on a left ventricular pressure wave. The left ventricular end-diastolic pressure should be measured at end expiration. In this case, the left ventricular end-diastolic pressure is nearly 50 mm Hg.

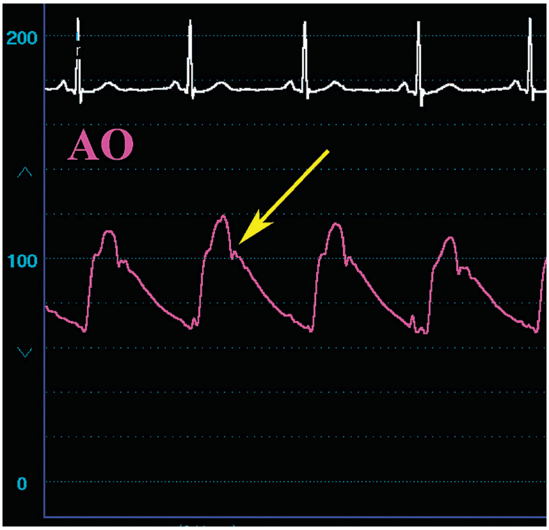


Fig. 2.23 An example of a normal, high-fidelity aortic (AO) pressure waveform. Note the dicrotic notch *(arrow).*

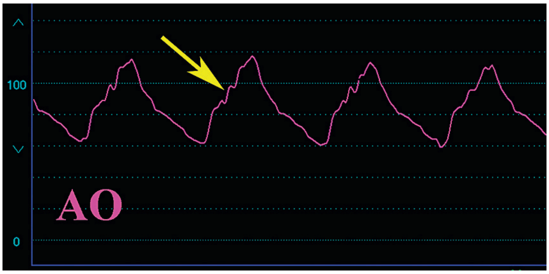
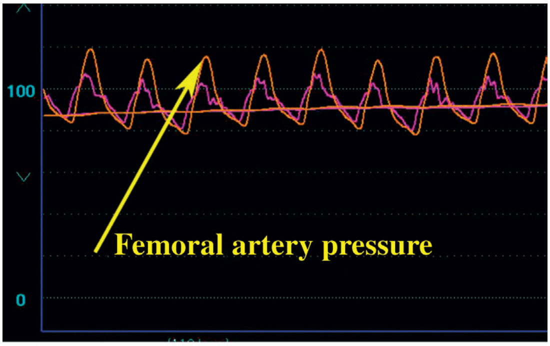


Fig. 2.24 This waveform demonstrates an example of a prominent anacrotic "notch" or "shoulder" *(arrow)* in a patient with severe aortic (AO) regurgitation. An anacrotic notch may also be observed in patients with severe AO stenosis and indicates turbulence during ejection.



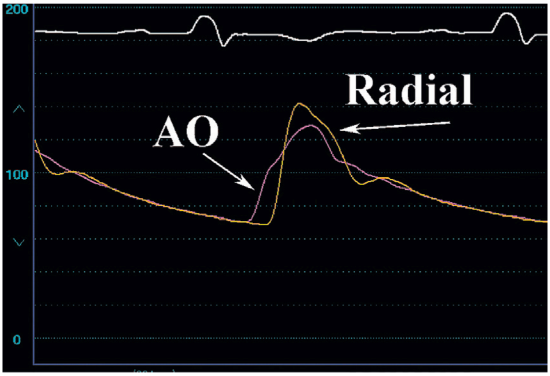


Fig. 2.25 (A) A simultaneous central aortic (AO) pressure and femoral artery sheath pressure obtained in a patient with AO stenosis. Compared with the central AO pressure, the femoral artery pressure wave exhibits a time delay, peripheral amplification with higher systolic pressure, and a "damped" appearance with loss of the dicrotic notch *(arrow)*. (B) A simultaneous central AO and radial artery pressure showing the typical "spiked" appearance of a peripheral waveform.

The effect of reflected waves is particularly notable in peripheral arterial waveforms (brachial, femoral, and radial) (Fig. 2.25). Peak systolic pressure exceeds central aortic pressure by 10 to 20 mm Hg due to peripheral amplification from reflected waves. The contour of the waveform changes as the pressure is sampled further from the aortic valve, with a steeper upstroke, narrower systolic portion (spiked appearance), and markedly diminished or absent dicrotic notch.

MEASUREMENT OF SIMULTANEOUS PRESSURES IN TWO OR MORE CHAMBERS

Cardiac catheterization protocols collect simultaneous pressure in two or more chambers for the purpose of screening for several specific conditions. Simultaneous left ventricular and right ventricular pressures and simultaneous left ventricular and PCWP recordings should be obtained as part of a complete hemodynamic study that involves a right and left heart catheterization. The purpose of these two maneuvers is to screen for the presence of occult restrictive/constrictive physiology and mitral stenosis, respectively. Additional simultaneous pressure measurements include the collection of left ventricular and central aortic pressure for the diagnosis of aortic stenosis, left ventricular outflow tract obstruction, or coarctation of the aorta, and simultaneous right ventricular and pulmonary artery pressures collected if there is pulmonic valve or pulmonary artery stenosis. Simultaneous left atrial and left ventricular pressures are performed to assess for mitral stenosis when a PCWP is not adequate or possible.

COMBINATION OF LEFT VENTRICULAR AND RIGHT VENTRICULAR PRESSURES

This maneuver is performed to screen for the presence of restrictive or constrictive physiology and should be a part of all complete hemodynamic evaluations. Fairly complex interactions occur between the cardiac chambers, the pericardium, and the intrathoracic cavity during the respiratory cycle. In general, in patients without constrictive physiology, the net effect of these interactions causes left ventricular and right ventricular diastolic pressures to differ by at least 5 mm Hg, with variation during the respiratory cycle. Peak systolic pressure changes during inspiration and expiration in the right and left ventricles parallel each other (Fig. 2.26). Abnormalities in these relationships are observed in constrictive pericarditis and restrictive cardiomyopathy. A detailed discussion of these interactions and the hemodynamic effects of pericardial and restrictive myocardial diseases is the subject of Chapter 10.

The hemodynamic effect of conduction abnormalities is often reflected in simultaneous right ventricular and left ventricular pressure tracings. Normally, the right ventricular waveform sits within the confines of the left ventricular wave (Fig. 2.27A). A right bundle branch block delays the right ventricular contraction relative to the left ventricular contraction, resulting in a delay in the right ventricular pressure wave and a shift to the right (see Fig. 2.27B). The opposite occurs with a left bundle branch block (see Fig. 2.27C).

COMBINATION OF PULMONARY CAPILLARY WEDGE (OR LEFT ATRIAL) AND LEFT VENTRICULAR PRESSURE

The purpose of this maneuver is to screen for the presence of mitral valve stenosis. In addition, simultaneous left ventricular pressure-PCWP is needed to determine the mitral valve orifice area in patients with known mitral stenosis. Normally, no gradient should exist between PCWP and LVEDP (Fig. 2.28). A small gradient may be discernible only early in diastole when the left ventricular diastolic pressure is lowest. This small gradient naturally allows blood to flow from the left atrium to the left ventricle. By the end of diastole, left atrial and left ventricular pressures should be the same, unless there is mitral stenosis. The time delay associated with the PCWP tracing causes the *a* and *v* waves to appear later in the left ventricular waveform. This may be problematic if a large *v* wave exists because it may be confused with a diastolic gradient (Fig. 2.29). However, note the location of the *v* wave on a true left atrial waveform obtained by transseptal catheterization (Fig. 2.30). The true position of the *v* wave as seen on the left atrial waveform occurs during ventricular systole, and the *y* descent correlates with the decay in the left ventricular pressure. A large *v* wave on the PCWP, coupled with the time delay, gives the false impression that there is a diastolic gradient; this can be corrected if desired by shifting the PCWP to the left, allowing proper alignment with the left ventricular waveform.

COMBINATION OF CENTRAL AORTIC AND LEFT VENTRICULAR PRESSURE

Simultaneous left ventricular and central aortic pressures are recorded in cases of known or suspected aortic stenosis as a method of determining the transvalvular gradient necessary to estimate valve area. Simultaneous left ventricular and femoral arterial sheath pressures have been used in the past for this purpose, but discrepancies between central aortic and femoral arterial sheath pressures are commonly observed. The previously discussed phenomenon of peripheral amplification results in a higher sheath than central aortic pressure (see Fig. 2.25A). In contrast, peripheral vascular disease, catheter thrombosis, or kinking of the sheath cause a higher central aortic than sheath pressure. For these reasons, it is best to use simultaneous left ventricular and central aortic pressure for this purpose, and additional details regarding this assessment will be provided in Chapter 6.

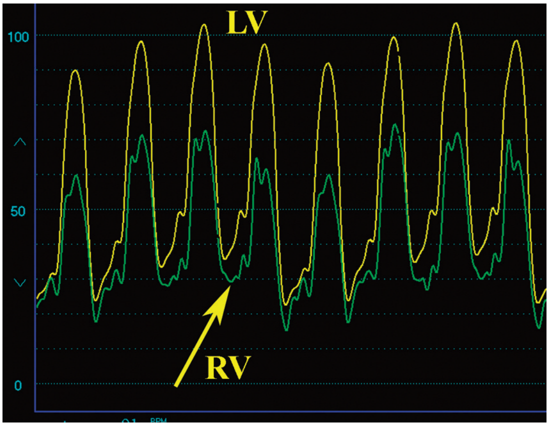


Fig. 2.26 Simultaneous left ventricular (LV) and right ventricular (RV) waveforms in a patient with heart failure and pulmonary hypertension, showing the normal relationship between the two chambers. The systolic pressures rise and fall together during the respiratory cycle, and the diastolic pressures are more than 5 mm Hg apart *(arrow)*.

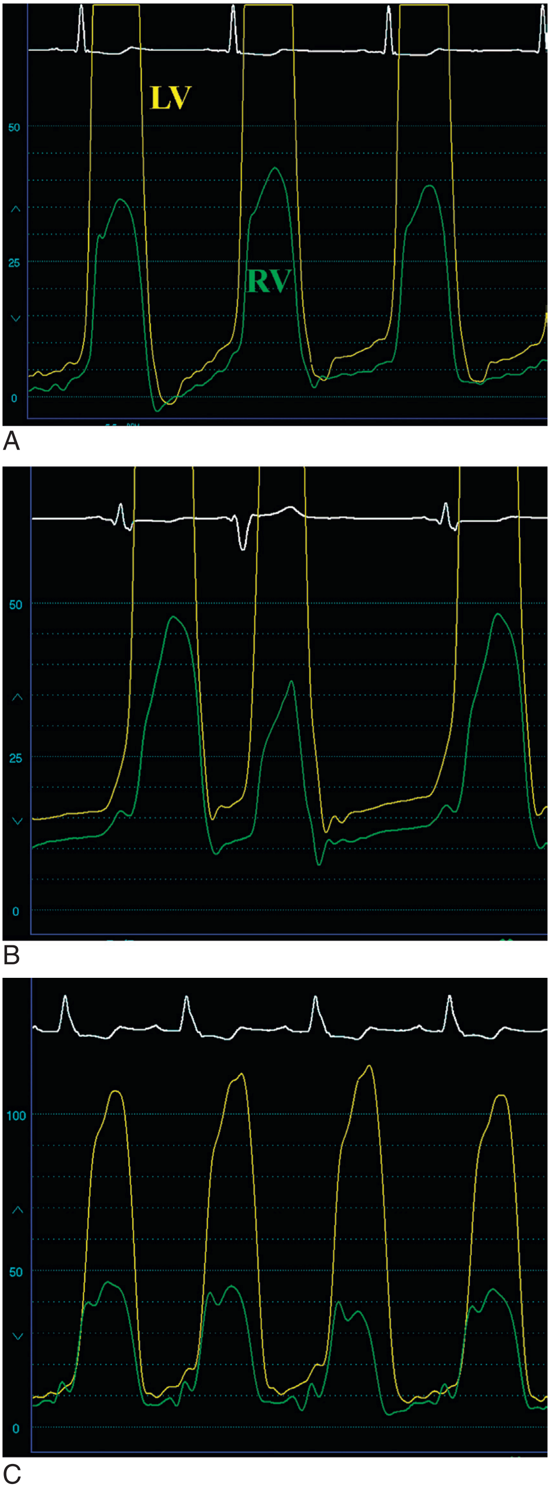


Fig. 2.27 Effect of conduction abnormalities on simultaneous right ventricular (RV) and left ventricular (LV) pressure waveforms. (A) Normally, the RV pressure wave sits within the LV pressure contour. (B) In the presence of a right bundle branch block, the RV wave is shifted to the right. Note how a premature ventricular contraction returns the RV tracing to a more normal appearance. (C) A left bundle branch block or intraventricular conduction delay interrupts the LV contraction relative to the right ventricle, causing the RV waveform to shift to the left.

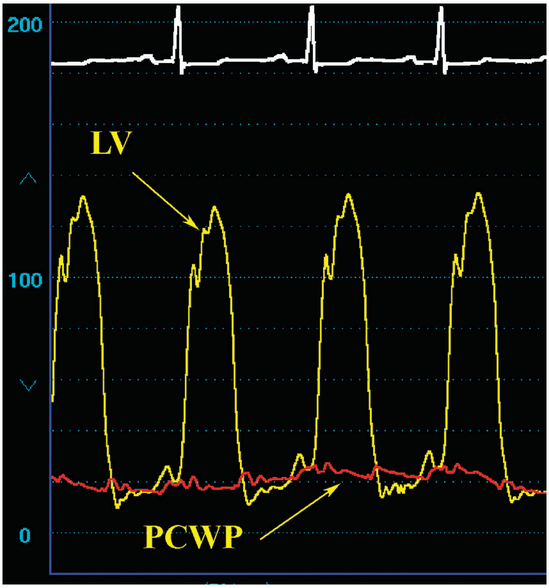


Fig. 2.28 Simultaneous left ventricular (LV) and pulmonary capillary wedge pressure (PCWP) tracings obtained during a routine cardiac catheterization protocol as a screening test for mitral stenosis. As shown here, no pressure gradient normally exists late in diastole between the two chambers.

Fig. 2.29 Simultaneous left ventricular and pulmonary capillary wedge pressure tracings obtained in a patient with severe mitral regurgitation, demonstrating a very large *v* wave. The time delay inherent to the wedge pressure waveform suggests the presence of a mitral gradient during diastole. Phase shifting the wedge tracing to the left corrects this false gradient.

COMMON ERRORS AND ARTIFACTS

Most errors in the collection and interpretation of hemodynamic data are related to one of the reasons listed in Box 2.2. Erroneous data due to an improper zero level or unbalanced transducer might lead to patient mismanagement. With the commonly used fluid-filled systems, artifacts and errors may be caused by a small air bubble, catheter kink, or blood clot anywhere along the line from the catheter tip to the sensing membrane. Similarly, a loose connection or stopcock

Fig. 2.30 This simultaneous left ventricular (LV) and left atrial (LA) pressure tracing was obtained in a patient with significant mitral regurgitation using a transseptal approach to measure the LA pressure. A prominent *v* wave is apparent, but no mitral gradient is present. Note the timing of the *v* wave on the LA pressure tracing relative to the LV pressure waveform. As shown in Fig. 2.29, the time delay inherent to pulmonary capillary wedge pressure recordings causes the *v* wave of a pulmonary capillary wedge pressure to appear later in diastole and may lead to a false diagnosis of a mitral gradient.

Box 2.2 Common Sources of Error or Inaccuracy in Hemodynamic Assessment

1. Improper zero level or transducer balancing
2. Air bubbles, clots, or kinks in the system
3. Loose connections
4. Defective transducers
5. Tachycardia and loss of frequency response
6. Mechanical ventilators and excessive intrathoracic pressure changes
7. Artifacts:
   Overdamping
   Overshoot or “ring” artifact
   Catheter whip or “fling”
   Catheter entrapment
   Hybrid waveforms

can cause inaccuracy. Transducers may be defective or poorly calibrated, which should be considered when there is difficulty maintaining transducer balance or if the data obtained appear inconsistent. The frequency response of most clinically used systems may be exceeded in the presence of marked tachycardia, preventing the collection of high-fidelity tracings. Mechanical ventilators and extreme changes in intrathoracic pressure can make interpretation difficult. Finally, several artifacts frequently thwart an unsuspecting clinician.

Probably the most commonly observed artifacts relate to an improper degree of damping. The overdamped tracing (Fig. 2.31) indicates the presence of excessive friction, absorbing the force of the pressure wave somewhere in the line from the catheter tip to the transducer. The tracing lacks proper fidelity and appears smooth and rounded because of the loss of frequency response. This will result in the loss of data and will falsely lower peak pressures. Typically, the dicrotic notch on the aortic or pulmonary artery waveforms is absent, and the right atrial or PCWP waveforms will lack distinct *a* and *v* waves. The diastolic pressure tracing on an overdamped ventricular waveform will be smooth, preventing the recognition of *a* wave and making determination of end-diastolic pressure difficult. This artifact is usually caused by sloppy operators with air bubbles in the tubing, catheter, or transducer or a loose connection anywhere in the system. The operator should also be aware that a thrombus or kink in the catheter may also cause this artifact, as well as the presence of high-viscosity radiographic contrast agents in the catheter. For the latter reason, hemodynamic data should always be collected with saline, and not contrast, in the catheter.

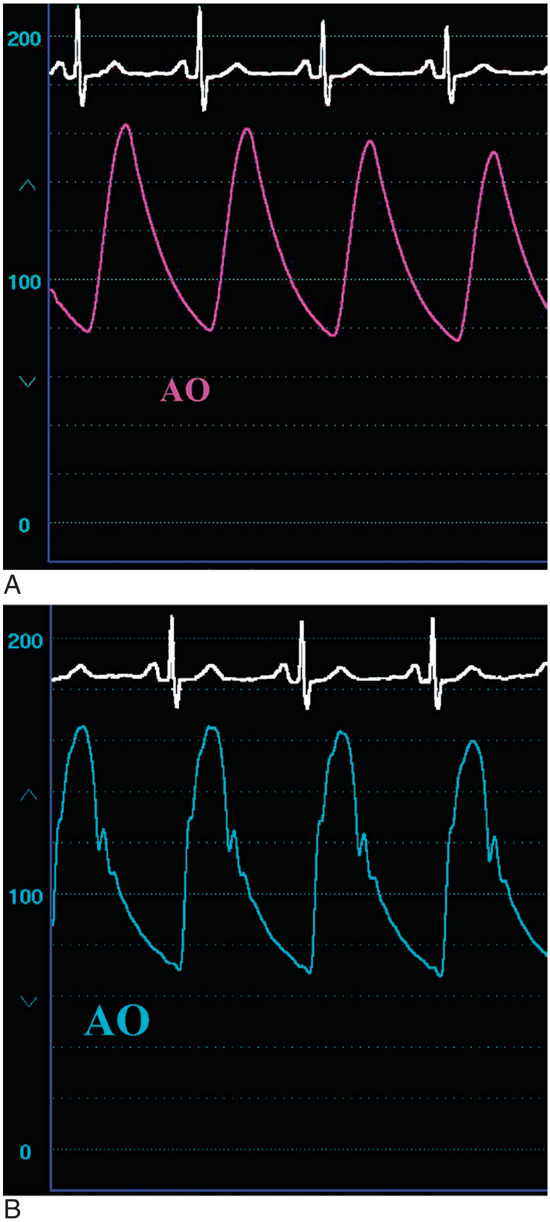


Fig. 2.31 (A) A damped aortic (AO) pressure waveform due to the presence of an air bubble in the catheter. This tracing has poor fidelity, a smooth appearance, and lacks a dicrotic notch. (B) Following vigorous flushing and removal of the offending air bubble, a high-fidelity tracing is apparent with the return of a dicrotic notch.

Underdamping causes *overshoot* or *ring artifact* (Figs. 2.32 and 2.33). This artifact typically appears as one or more narrow "spikes" overshooting the true pressure during the systolic pressure rise with similar, negatively directed waves overshooting the true pressure contour during the downstroke. This artifact may lead to overestimation of the peak pressure and underestimation of the pressure nadir. Tiny air bubbles that oscillate rapidly back and forth, transmitting energy back to the transducer, often cause this artifact. Flushing the catheter or transducer often corrects this

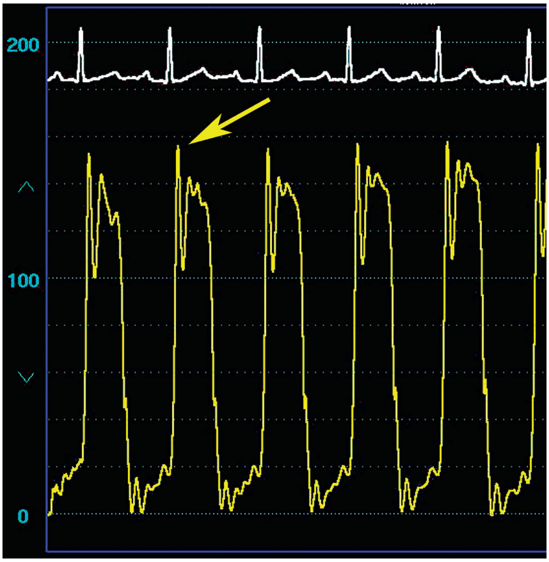


Fig. 2.32 "Ring" artifact is a very common artifact, usually due to the presence of a small bubble somewhere in the system between the catheter tip and the transducer. The small bubble oscillates, causing the high-frequency, spiked artifact shown here *(arrow)*. This can usually be corrected by flushing the catheter or introducing a filter.

artifact; alternatively, the introduction of a filter to the hemodynamic system may be necessary to eliminate this artifact. Fig. 2.33 is an example of an overshoot artifact on a right ventricular waveform, resulting in a systolic pressure higher than the pulmonary artery pressure. If unrecognized, this might be falsely diagnosed as pulmonic stenosis.

Related to the overshoot or ring artifact is *catheter whip* or *fling artifact* (Fig. 2.34). This artifact is created by the acceleration of the fluid within the catheter from rapid catheter motion and is commonly seen with balloon-tipped catheters in hyperdynamic hearts or balloon-tipped catheters placed in the pulmonary artery with extraneous loops. Similar to ring artifact, catheter whip causes overestimation of the systolic pressure and underestimation of the diastolic pressure. This artifact is difficult to remedy; eliminating the extra loops or deflation of the balloon can improve the appearance and limit this artifact.

Catheter malposition creates several interesting artifacts. *Catheter entrapment artifact* is sometimes observed, particularly when measuring left ventricular pressure with an end-hole catheter. Cardiac catheterization in patients with hypertrophic obstructive cardiomyopathy entails a search for an intraventricular pressure gradient. An end-hole catheter allows the operator to determine the precise location of a pressure gradient. Unfortunately, during these efforts the tip of the catheter may become buried or "entrapped" within the hypertrophied myocardium and reflect intramural rather than intracavitary pressure. The resulting bizarre, spiked appearance to the left ventricular waveform may lead to a false diagnosis of a left ventricular outflow tract gradient (Fig. 2.35). A *hybrid tracing* results when the sampled pressure represents a mixture of the waveforms from more than one cardiac chamber. Hybrid tracings are observed in two common clinical scenarios. In one scenario, a catheter such as a pigtail catheter, consisting of multiple side holes, straddles two cardiac chambers. The pressure waveform conveyed to the transducer contains pressure elements from each chamber. For example, a pigtail catheter may be improperly positioned within the left ventricle, with side holes lying above and below the aortic valve. The pressure waveform will contain both aortic and left ventricular pressure waveform elements, creating a hybrid of both chambers (Fig. 2.36). Hybrid tracings may also be observed during attempts at obtaining a PCWP, particularly if there is pulmonary hypertension. In such cases the catheter may not completely occlude the pulmonary artery; the resulting waveform is contaminated by the pulmonary artery pressure and represents a mixture of the pulmonary artery and pulmonary capillary waveforms, falsely elevating the wedge pressure. This artifact may be responsible for a false diagnosis of decompensated heart failure or mitral stenosis in patients with pulmonary hypertension. It may be difficult to detect because, in fact, characteristic *a* and *v* waves may be observed (Fig. 2.37). As noted earlier, confirmation of the true PCWP position will require measurement of blood oximetry in such cases.

Finally, catheter malposition against a heart valve or wall of a blood vessel or cardiac chamber may create damping of the waveform when the catheter tip lies against a chamber wall (usually a low-pressure chamber such as the right atrium) or an abnormal "spike" artifact when the catheter tip strikes a heart valve. This may be misinterpreted as a *v* wave on a right atrial waveform (Fig. 2.38).

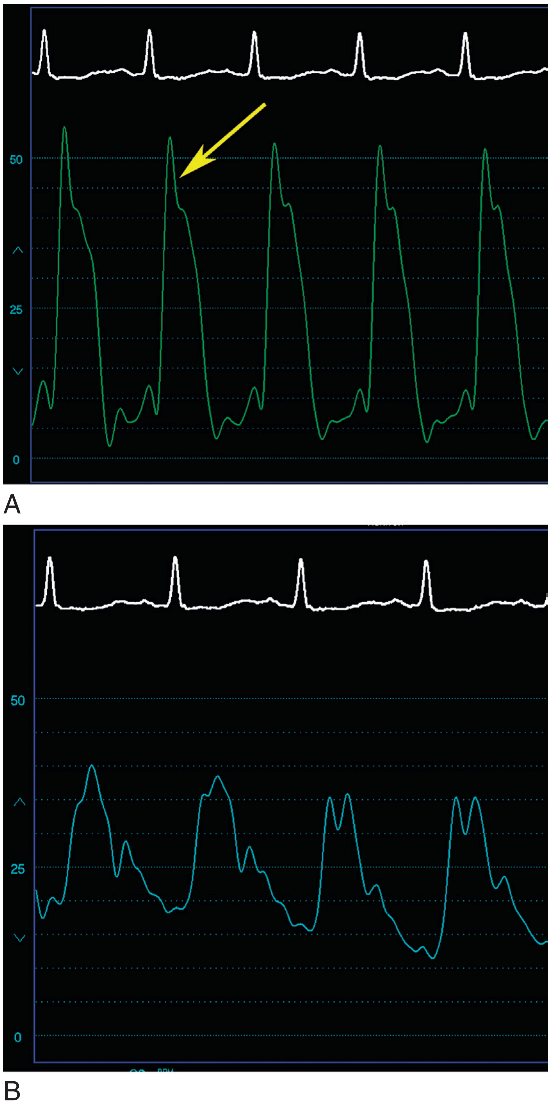
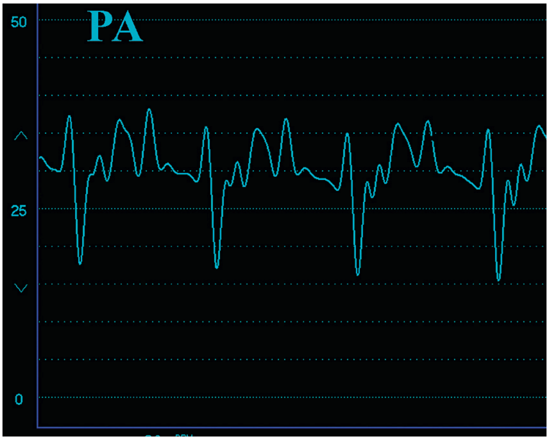


Fig. 2.33 A commonly seen “overshoot” artifact in a right ventricular pressure tracing. (A) The overshoot portion *(arrow)*. (B) This may provide the false impression that the right ventricular pressure exceeds the pulmonary artery pressure, leading to an erroneous diagnosis of pulmonary artery or pulmonic valve stenosis.

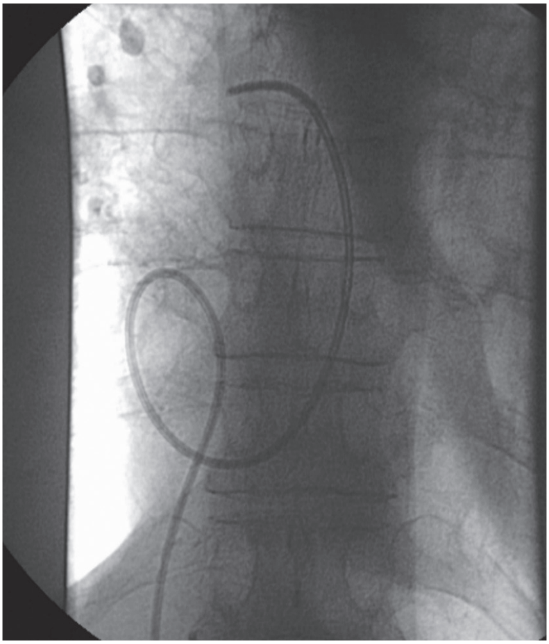
EFFECT OF ARRHYTHMIA ON HEMODYNAMIC MEASUREMENTS

The cardiac rhythm determines the events responsible for the generation of the pressure waveforms during the cardiac cycle, and disturbances of the cardiac rhythm are often reflected in the waveforms. Perhaps the simplest and most commonly observed rhythm abnormality is the premature ventricular contraction (PVC). The premature beat produces a systole with a contraction that may be too weak to open the aortic valve. This may be apparent hemodynamically in the absence of the arterial pressure wave. This causes the “skipped beat” reported by many patients (Fig. 2.39A). More often, however, the premature beat is associated with reduced aortic pressure (see Fig. 2.39B). Because of the lower pressures of the right heart, the right ventricular contraction associated with a premature beat is usually adequate to open the pulmonic valve and generate a weakened pulmonary artery pressure wave. This does create the scenario, however, in which one ventricle ejects (the right ventricle), while the other does not (the left ventricle) and may explain the symptoms of "fullness" and dyspnea accompanying PVCs in some patients.

A normal beat after PVC follows a compensatory pause and is typically a beat with a stronger contraction due to enhanced filling and increased contractility.[15] This may manifest as an increased systolic pressure on the aortic or left ventricular pressure wave compared with the usual sinus beats (Fig. 2.40). Most normal hearts do not exhibit this behavior, and it is more readily apparent in patients with left ventricular dysfunction in whom the greater contractile force of the post-PVC has more impact on the cardiac output. The phenomenon of extrasystolic potentiation assists in the diagnosis of hypertrophic obstructive cardiomyopathy and will be fully described in Chapter 7. In brief, the increased contractility of the

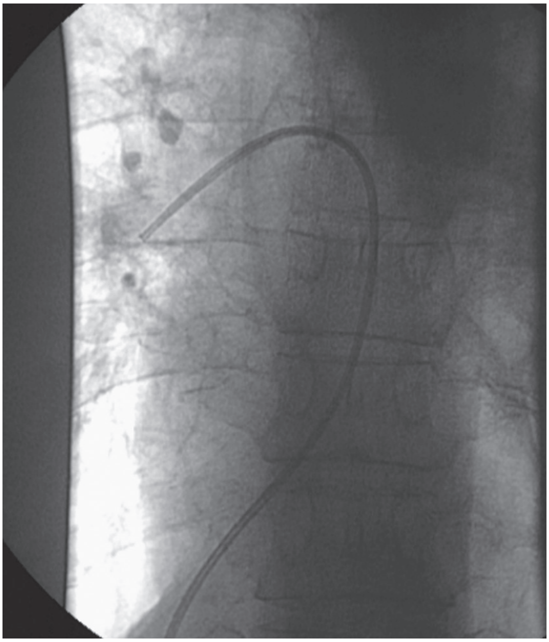


A



B

Fig. 2.34 Catheter "fling" or "whip" is another common artifact usually seen in the pulmonary artery (PA) pressure waveform in patients with hyperdynamic hearts. (A) A PA pressure waveform was nearly unrecognizable. (B) This was due to excessive catheter whip from a prominent loop in the right atrium.



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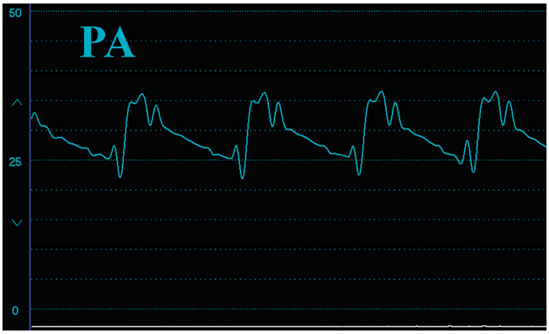

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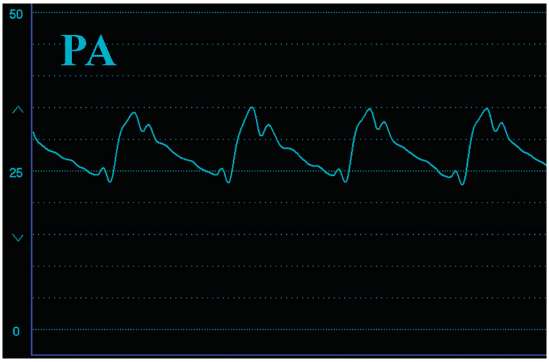

E

Fig. 2.34, cont'd (C) Removal of the loop improved the appearance of the waveform. (D) Overshoot artifact still remained. (E) Addition of a filter resulted in further improvement in the appearance of the waveform.

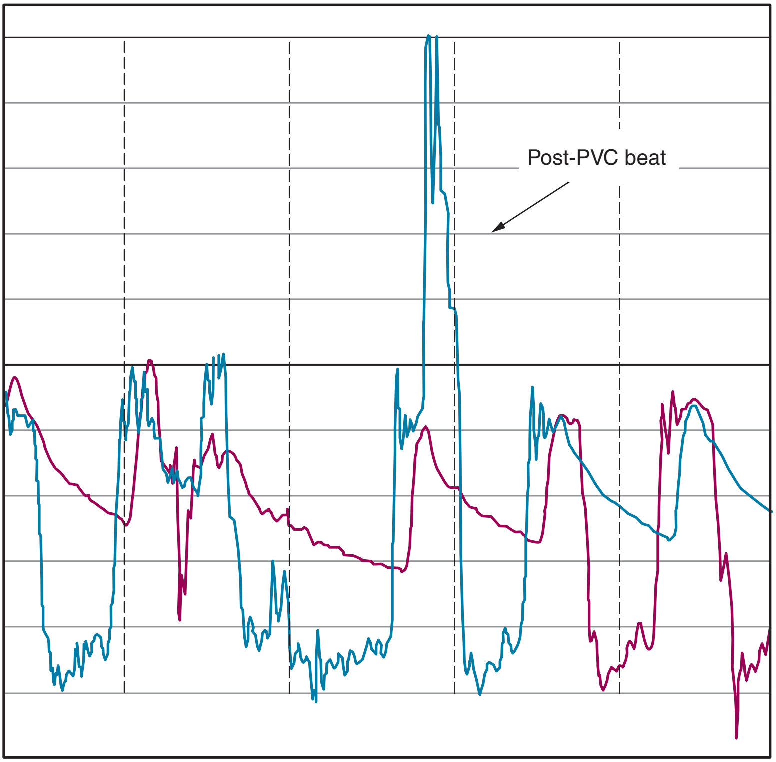

Fig. 2.35 Catheter entrapment artifact in a patient with marked left ventricular hypertrophy undergoing catheterization to determine the presence of a left ventricular outflow tract gradient. Note that the beginning phase of ventricular systole in the post-premature ventricular beat (PVC) appears similar to other beats but is then followed by a bizarre, spiked deflection representing intramyocardial pressure due to catheter entrapment.

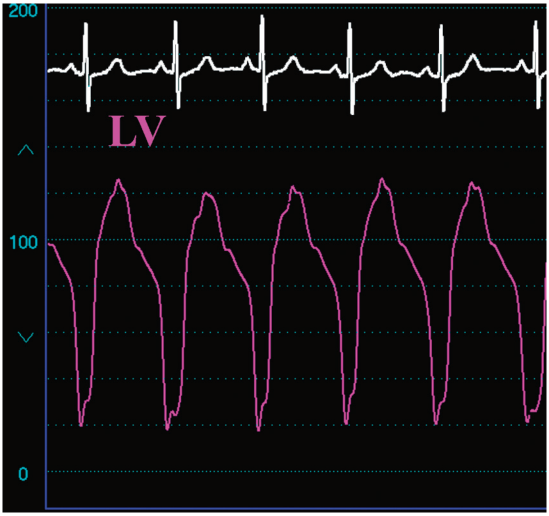

Fig. 2.36 A hybrid waveform obtained from a pigtail catheter placed in the left ventricle. Some of the side holes of the catheter are in the aorta, causing this peculiar waveform. More subtle versions of this artifact may not be recognized and lead the observer to falsely assume elevation of the left ventricular (LV) end-diastolic pressure.

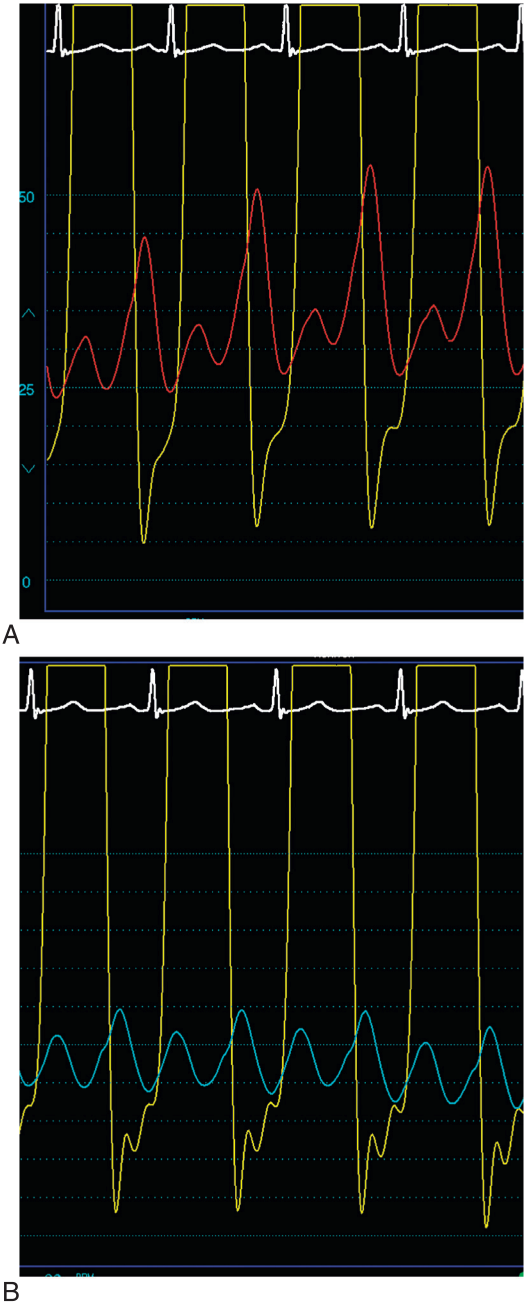

Fig. 2.37 Hybrid tracings are very common during attempts to measure pulmonary capillary wedge pressure. In this example, elevated pulmonary artery pressure is present and measures 61/30 mm Hg. (A) With a balloon flotation catheter, the "wedge" pressure reveals distinct *a* and *v* waves and suggests the presence of a mitral valve gradient. However, the oxygen saturation of the blood withdrawn from the catheter tip measures 72%. (B) With the catheter positioned more distally, a better pulmonary capillary wedge pressure tracing is obtained with an oxygen saturation of 95%, thus confirming a true "wedge" position and the absence of a mitral valve gradient.

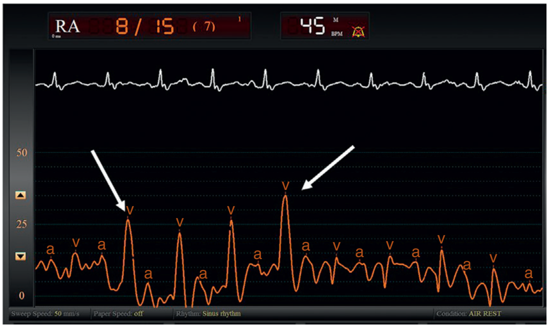

Fig. 2.38 This is an example of an artifact commonly observed in the right atrial waveform. The catheter tip is struck by the tricuspid valve and generates a waveform misinterpreted as a *v* wave *(arrows)*. This is an artifact from catheter malposition. *RA*, Right atrium.

Fig. 2.39 An example of an aortic pressure waveform obtained in a patient with ventricular bigeminy. (A) The premature ventricular beats *(arrows)* are unable to generate enough force to open the aortic valve, and there is no associated aortic pressure waveform. (B) Each premature beat is associated with a diminished aortic pressure *(arrow)*.

post-PVC increases the degree of obstruction, raising the pressure gradient between the aorta and the left ventricle but lowering the pulse pressure on the aortic pressure waveform (Fig. 2.41). Compare this with the patient with valvular aortic stenosis, where the degree of obstruction is fixed and the post-PVC raises the gradient but also raises both left ventricular and aortic systolic pressure; the pulse pressure on the aortic waveform consequently increases (Fig. 2.42).

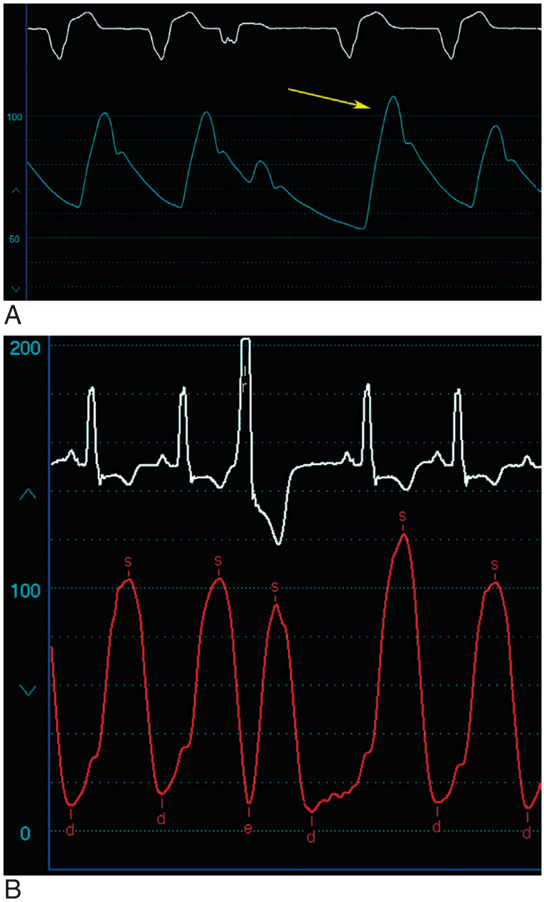

Fig. 2.40 The beat following a premature ventricular contraction typically has greater contractile force and may have a higher systolic pressure *(arrow)*. (A) Depiction of finding on aortic pressure trace. (B) Depiction of finding on a left ventricular tracing. *d*, Diastole; *e*, end diastole; *s*, systole.

Atrial premature beats may be difficult to appreciate on the ECG but significantly affect the hemodynamic waveforms and may lead to errors in interpretation. An interesting example is shown in Fig. 2.43, where the aortic pressure appears to show pulsus alternans. On close inspection of the monitoring lead, however, an atrial bigeminal pattern is apparent. This finding has deceived experienced cardiologists since the 1960s, when it was also described with sinus arrhythmia.[16]

Atrial fibrillation is a very common dysrhythmia with important effects on the hemodynamics. On the right atrial and left atrial waveforms, the absence of atrial systole is noted by the absence of an *a* wave. The *x* descent, however, may still be evident because this is caused by downward excursion of the tricuspid or mitral annulus during ventricular systole (Fig. 2.44). In atrial flutter, atrial systole occurs, albeit at a very rapid rate, and this translates to clearly notable systolic waves on the atrial pressure tracing (Fig. 2.45). The varying RR intervals in atrial fibrillation cause varying degrees of ventricular filling with each beat and result in a wide variation in the systolic pressure peaks (Fig. 2.46). This may be confused as a pulsus paradoxus. Furthermore, this same phenomenon results in beat-to-beat variations in the transvalvular gradients that impact valve area calculations in patients with aortic or mitral stenosis (Figs. 2.46 and 2.47). It is for this reason that data from at least 10 consecutive beats are averaged in patients with atrial fibrillation when calculating the valve area invasively.

Complete heart block results in the loss of AV synchrony. Contraction of the atrium against a closed valve causes a marked increase in atrial pressure, particularly if this chamber is not compliant. This can be appreciated on a right atrial waveform as a *cannon wave* and may be confused with prominent *v* waves, leading to a false diagnosis of tricuspid

Fig. 2.41 In patients with hypertrophic obstructive cardiomyopathy, the beat following a premature ventricular contraction is associated with greater obstruction, an increase in the outflow tract gradient, and a drop in aortic pulse pressure known as the Brockenbrough sign.

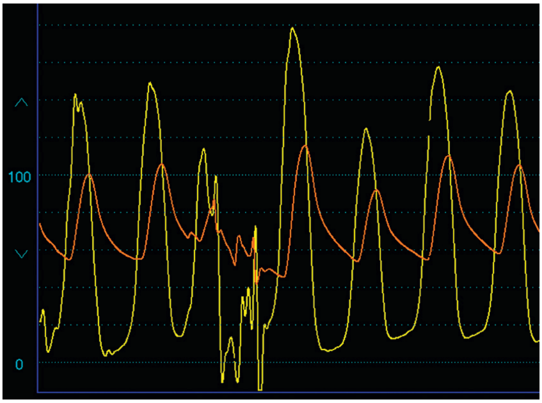

Fig. 2.42 In contrast to patients with hypertrophic obstructive cardiomyopathy, the augmented post-premature beat results in an increase in the transvalvular gradient and an increase in aortic pulse pressure in patients with valvular aortic stenosis.

regurgitation (Fig. 2.48). The importance of AV synchrony in patients with reduced left ventricular function and a ventricular pacemaker can be appreciated by the increase in systolic pressure when the atria contracts compared with beats where there is just ventricular contraction (Fig. 2.49). Additional hemodynamic abnormalities caused by pacemakers have been reviewed.[17] First-degree AV block causes the *c* wave to become more obvious since the *c* wave follows the *a* wave at the same time as the PR interval on the ECG (Fig. 2.50).

EFFECT OF OBESITY ON HEMODYNAMIC MEASUREMENTS

Obtaining accurate hemodynamic measurements in morbidly obese patients can be challenging. The *zero* level is often erroneously set to the same level as more Svelte patients, causing erroneous measurements of diastolic pressures. Importantly, patients with morbid obesity generate tremendous negative inspiratory forces during respiratory efforts and most prominently distort right heart pressure tracings. As noted earlier, the end-expiratory values should be used in such cases (Fig. 2.51). A pulsus paradoxus becomes more prominent and is often erroneously attributed to other

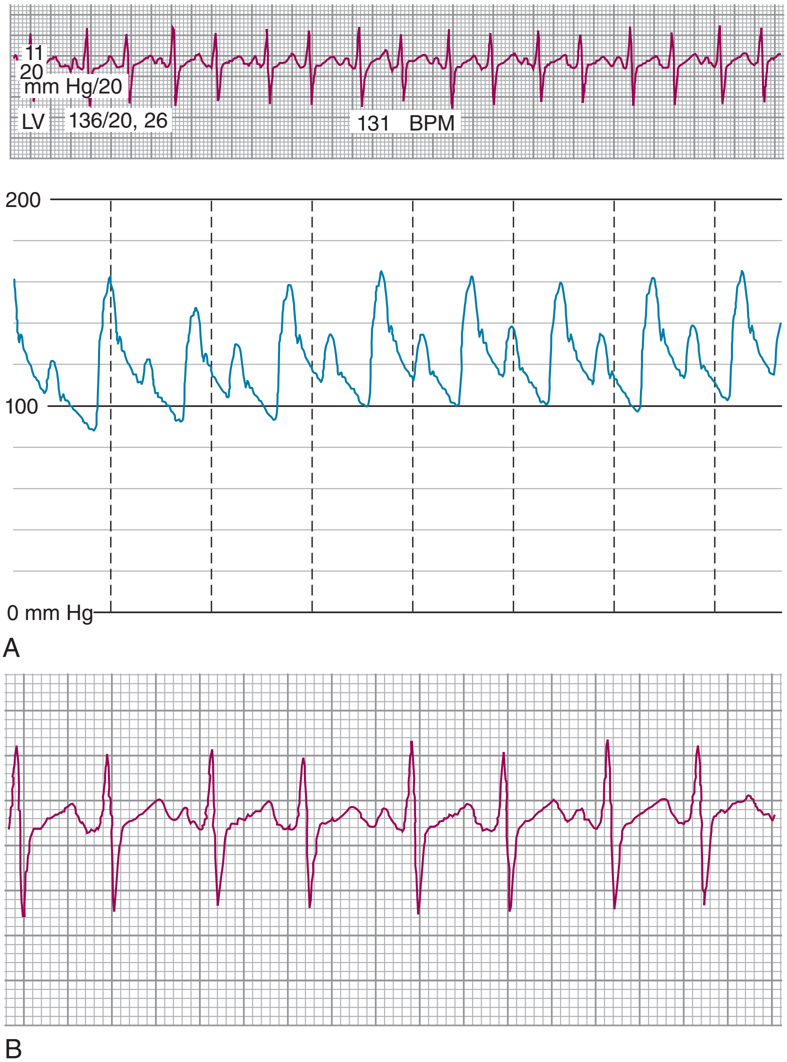

Fig. 2.43 (A) On initial observation, the aortic waveform on this tracing appears to show pulsus alternans. (B) Close inspection of the rhythm strip reveals atrial bigeminy. *BPM*, Beats per minute; *LV*, left ventricular.

Fig. 2.44 Atrial fibrillation distorts the atrial pressure waveform. Because atrial systole is not present, the wave is absent. However, an *x* descent may be seen due to downward displacement of the mitral or tricuspid annulus *(arrow)*.

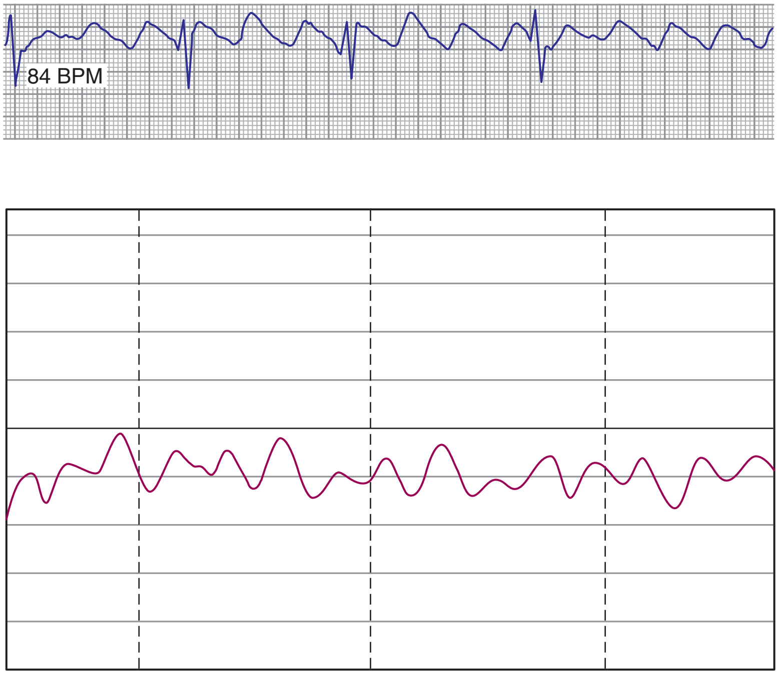

Fig. 2.45 In contrast to atrial fibrillation, atrial systoles are generated with atrial flutter, as shown. *BPM*, Beats per minute.

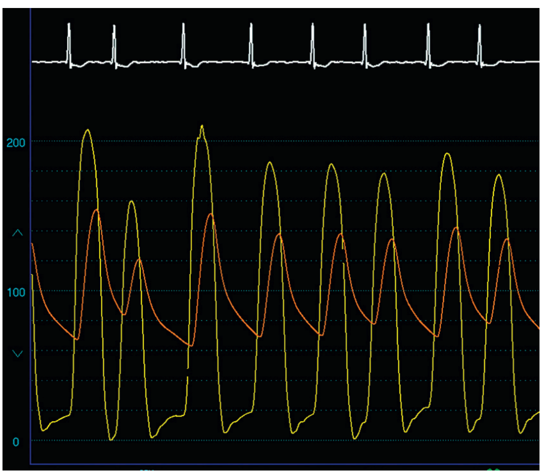

Fig. 2.46 In patients with atrial fibrillation and aortic stenosis, the varying RR intervals will result in different pressure gradients with each beat. In addition, systolic pressure will vary and may lead to a false diagnosis of pulsus paradoxus.

conditions, such as pericardial effusion. The right atrial, right ventricular diastolic, and pulmonary artery pressures may be modestly elevated in severe obesity. One study found approximately one-third of obese individuals to have pulmonary artery systolic pressures exceeding 35 mm Hg.[18] In addition, obesity is commonly associated with obstructive sleep apnea, a condition that also causes pulmonary hypertension.[19]

Fig. 2.47 The varying RR intervals seen in a patient with atrial fibrillation and mitral stenosis will result in different transvalvular pressure gradients. Long RR intervals will reach diastasis, whereas short intervals are associated with the greatest gradient. *d*, Diastole; *e*, end diastole; *s*, systole.

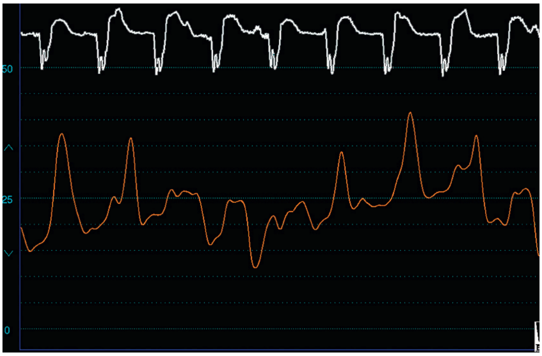

Fig. 2.48 Cannon *a* waves associated with complete atrioventricular block in a patient with a pacemaker.

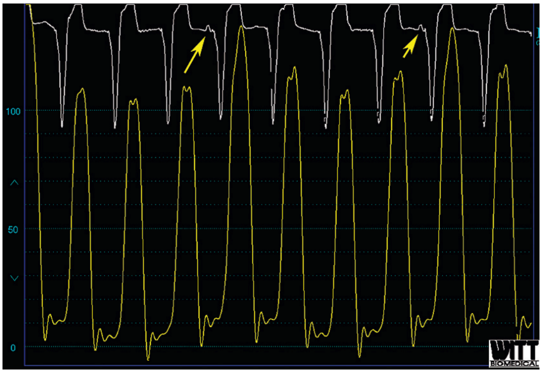

Fig. 2.49 This patient with known severe left ventricular dysfunction has a permanent pacemaker with only a ventricular lead and underlying severe sinus node dysfunction with complete heart block. Occasional sinus beats *(arrows)* allow for atrioventricular synchrony and result in increased cardiac output, as evidenced by the increase in systolic pressure.

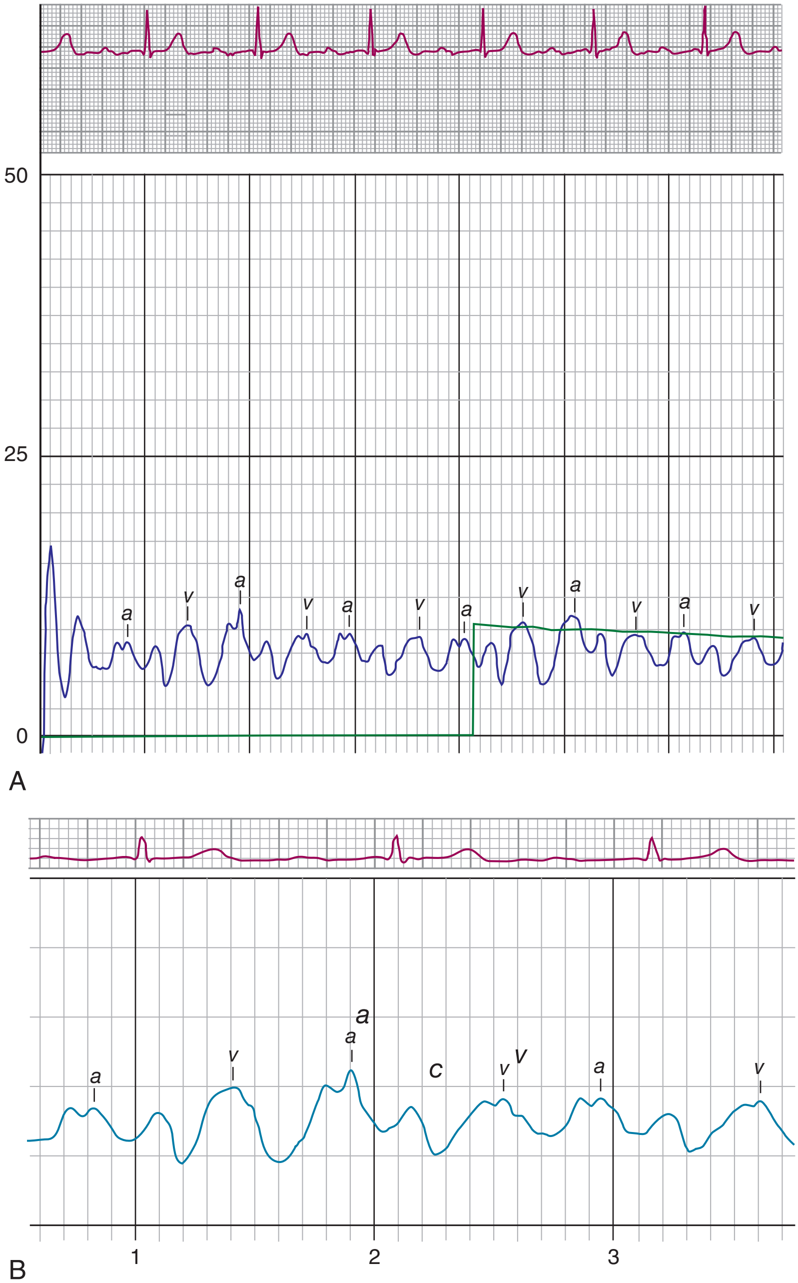

Fig. 2.50 (A) First-degree atrioventricular block accentuates the *c* wave on the right atrial waveform, as shown. (B) Individual waves are labeled.

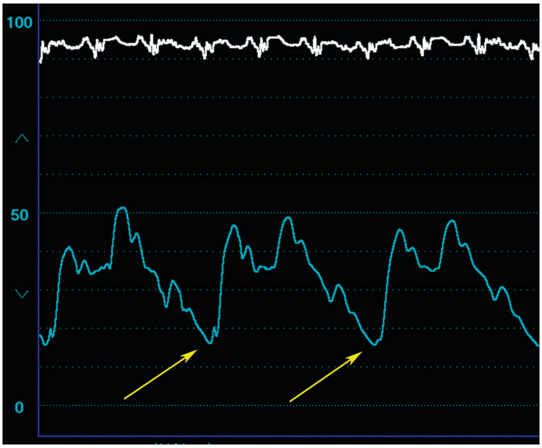

Fig. 2.51 Marked respiratory variation on a pulmonary artery pressure waveform in a patient with morbid obesity. The large negative inspiratory forces are apparent *(arrows)*.

EFFECT OF PULMONARY DISEASE ON HEMODYNAMICS

Pulmonary diseases may profoundly impact right heart hemodynamics. This has been understood for many years and, in fact, patients with chronic obstructive pulmonary disease were among the first individuals to undergo systematic right heart catheterization studies to further understand the effect of lung disease on cardiac hemodynamics.[20]

Patients with advanced lung disease have hyperinflated chests and may be unable to lie flat; these factors may cause challenges in determining the "zero" level. Alveolar hypoxia increases pulmonary vascular resistance. Initially, the reason is vasoconstriction, but, over time, elevations in pulmonary vascular resistance are caused by chronic, structural changes in the pulmonary vasculature. Pulmonary hypertension ensues and leads to right ventricular hypertrophy. Cor pulmonale is the term used to describe right heart failure on the basis of pulmonary diseas e and the associated pulmonary hypertension.

The hemodynamic abnormalities observed in chronic lung disease vary widely depending on the nature and extent of parenchymal damage and whether or not there is involvement of the pulmonary vasculature. Pure emphysema generally requires extensive parenchymal destruction before pulmonary pressures are elevated. The pulmonary artery pressures may be entirely normal despite advanced lung disease.[20] When pulmonary pressures are elevated, they generally reach modest levels that do not usually exceed systolic pressures of 40 to 50 mm Hg.[21] Right atrial and right ventricular diastolic pressures are typically in the normal range. Because of an associated increased inspiratory effort, there may be more prominent pressure changes due to respiration than normally observed, and a pulsus paradoxus greater than 12 to 15 mm Hg is not unusual. Patients who develop cor pulmonale, however, exhibit more profound hemodynamic abnormalities with marked elevations in pulmonary artery pressure. Elevations of both right atrial and right ventricular diastolic pressures reflect right heart failure, similar to those with pulmonary hypertension from other causes (see Chapter 9).

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